Reverse Engineering a Soft Landing*

Neville Francis

Jonghyuck Lim

Michael T. Owyang

October 29, 2026

Abstract

1 Introduction

Bringing inflation back to target without causing a recession is a central and longstanding challenge for monetary policy. A broad literature documents the output and employment costs of disinflation and measures these costs using historical episodes and alternative empirical models (Okun, 1978; Gordon and King, 1982; Ball, 1994; Cecchetti and Rich, 2001). These costs can depend importantly on nominal rigidities, expectations, and policy credibility (Fischer, 1984; Erceg and Levin, 2003). Historical evidence similarly suggests that large central-bank-induced disinflations have typically been accompanied by substantial economic sacrifice or recession (Cecchetti et al., 2023). A soft landing poses this tradeoff more sharply: Can inflation be returned to target without a recession?

The literature more directly focused on soft landings falls into two broad strands. A first strand uses historical evidence and model-based analysis to assess when soft landings occur and what conditions make them more or less feasible. Blinder (2023) reviews 11 Federal Reserve tightening episodes since 1965 and argues that landing outcomes depend on both

*The views expressed are those of the authors and do not necessarily reflect the views of the Federal Reserve Bank of St. Louis or the Federal Reserve System.

the conduct of monetary policy and the adverse external shocks. Using a broader international sample, Boissay et al. (2022) compare soft- and hard-landing episodes and find that unfavorable initial conditions and persistent supply disturbances increase the likelihood of a hard landing. Focusing on the subsequent easing phase that followed declines in elevated inflation, de Soyres and Saijid (2024) find that soft landings tend to occur when inflation was closer to target, growth was stronger, and the preceding tightening cycle was smaller. Domash and Summers (2022) argue from historical labor-market evidence that the combination of elevated inflation and exceptionally tight labor markets implied substantial risk of a hard landing. Taking a structural approach, Verbrugge and Zaman (2023) find that rapid disinflation without a recession becomes difficult when inflation is highly persistent, even under favorable assumptions, such as anchored inflation expectations and a rapid normalization of supply pressures.

A second strand uses policy counterfactuals to examine how alternative monetary-policy choices shape the tradeoff between disinflation and recessionary outcomes. In a policy application, Ajello et al. (2022) compare different speeds and degrees of monetary tightening and show that faster disinflation comes with substantially higher predicted recession probabilities. Hakamada and Walsh (2024) focus on policy delay, showing that postponing the response to an inflation shock worsens inflation outcomes. Using the ECB's estimated macroeconomic model, Pariès et al. (2024) evaluate alternative dates for the start of post-pandemic tightening and find that earlier lift-off would have reduced peak inflation but generated a prolonged recession, whereas delayed lift-off would have produced more persistent inflation. Bullard et al. (2026) study the recent U.S. soft landing in a model with potentially unanchored inflation expectations and find that the timing of tightening was critical for containing inflation without large real losses.

Relative to these studies, we pose the soft-landing problem in reverse. Rather than asking whether prevailing economic conditions permit a soft landing or what outcomes follow from a prespecified policy path, we condition on the desired outcome: an inflation target to be reached by a specified horizon and maintained through the forecast horizon, with an explicit ceiling on model-implied recession probability. For each historical starting condition, we recover the monetary policy intervention required to satisfy both restrictions.[1]

Our approach is closely related in spirit to the literature on forecast targeting and optimal monetary policy. In the forecast-targeting framework, monetary policy is organized around inflation and real activity projections, with the policy path chosen to bring those forecasts

[1]We focus on recessions rather than the more familiar output growth or unemployment outcomes for a few reasons. First, in constructing their recession dates, the NBER BCDC utilizes multiple measures of economic performance. Second, the recession target is independent of the history—that is, we assume the policymaker cares only about avoiding a recession not the magnitude of the drop in output growth.

into alignment with the central bank's objectives (Svensson, 1997; Svensson and Woodford, 2004; Woodford, 2007). Related empirical optimal-policy approaches use forecasts of policy objectives and estimated responses to monetary-policy shocks to evaluate or construct policy interventions under an explicit loss function (Barnichon and Mesters, 2023; McKay and Wolf, 2023). In the spirit of this literature, we organize the policy problem around desired future macroeconomic outcomes, but impose the objectives differently. Here, we impose the inflation target and recession-risk ceiling as explicit constraints and seek the smallest policy intervention that satisfies both.

Empirically, we combine a Qual-VAR (Dueker, 2005; McCracken et al., 2022) with a constrained optimization framework to determine minimum-effort monetary policy interventions. The Qual-VAR allows the binary NBER recession indicator to enter a multivariate time-series model through a latent recession factor. The forecast distribution of this latent variable provides the model-implied recession probabilities used in the counterfactual constraint. Within this system, we jointly identify two distinct monetary-policy shocks. A conventional monetary policy shock (MP) captures an unexpected change in the current policy rate, while an anticipated monetary policy shock (AMP) captures revisions to the expected near-term policy path while leaving the federal funds rate unchanged on impact. The two shocks are identified within the same orthogonal structural system using sign, narrative, and zero restrictions.

We implement the inverse experiment by solving a convex quadratic program that minimizes the sum of squared structural shocks subject to the policy objectives. Methodologically, this extends the structural scenario analysis of Antolín-Díaz et al. (2021), which imposes equality restrictions on future variables and shocks, to accommodate inequality restrictions such as a ceiling on recession risk. The benchmark requires inflation to reach 2 percent by month 24 and remain at that level through month 48, while keeping model-implied recession probability below 30 percent through month 48. May 1981 and July 2022 illustrate the policy-design mechanism. We then examine 24 historical starting dates selected to span observed combinations of inflation pressure, model-implied latent recession factor, and the federal funds rate.

We document three main findings. First, observed inflation at the starting date is an incomplete guide to how much disinflation policy must deliver. The relevant burden is the additional inflation reduction left for policy to deliver after accounting for the underlying inflation outlook. May 1981 and July 2022 begin with similarly high 12-month CPI inflation, but their subsequent inflation dynamics differ substantially. The unconditional forecast leaves 4.8 percentage points of required policy disinflation in May 1981, more than twice the 2.2 percentage points required in July 2022. Across the representative historical starting

conditions, initial inflation and required policy disinflation are positively related, but the relationship is far from one-for-one, so current inflation alone does not rank the required intervention.

Second, incorporating recession risk as an explicit constraint materially changes the policy mix. When policy shocks are chosen solely to achieve the inflation objective, AMP accounts for about 30.0 percent of policy-induced disinflation in May 1981 and 31.6 percent in July 2022. Imposing the 30 percent recession-risk cap raises these shares to 75.3 and 46.6 percent, respectively. Across the representative starting dates, limiting recession risk increases both the AMP share and required policy effort. This pattern reflects a tradeoff between the two policy channels: MP shocks deliver greater disinflation per unit of shock but have stronger recessionary effects, while AMP shocks offer greater recession-risk control at the cost of larger policy interventions. Consistent with this tradeoff, AMP-only interventions have higher posterior feasibility shares than MP-only interventions when both objectives are imposed.

Finally, allowing more time for disinflation reduces policy effort but does not lower recession risk. Where the recession-risk cap binds, it continues to bind at longer target horizons. The minimum-effort policy uses the additional time to shrink the intervention and shift toward conventional policy, rather than to bring recession risk below the cap. Where the cap does not bind, peak recession probability also changes little, because the inflation reduction that policy must deliver barely changes with the horizon.

The remainder of the paper is organized as follows. Section 2 presents the empirical framework, structural identification, and soft-landing counterfactual design. Section 3 reports the counterfactual results, first for the two illustrative historical episodes and then across a broader set of representative historical starting conditions, including the policy mix and the effects of the disinflation horizon. Section 4 concludes.

2 Empirical Framework and Data

The empirical framework used to construct the policy scenarios has three layers: (i) the Qual-VAR estimates the macro-financial dynamics with a latent recession factor; (ii) structural identification decomposes reduced-form innovations into economically interpretable monetary policy shocks; and (iii) the scenario step iterates the identified SVAR forward and solves for the smallest future policy-shock path that satisfies a specified counterfactual scenario.[2]

[2]Appendix A collects the missing-data treatment, the sampler, the prior, the structural identification algorithm, and the solution of the scenario problem.

2.1 Qual-VAR with Latent Recession Factor

The empirical framework is a Qual-VAR that augments a monthly VAR with a Gaussian latent recession factor [see Dueker (2005) and McCracken et al. (2022)]. Let x_t denote the vector of observed macro-financial variables and S_t denote the binary NBER recession indicator. The recession indicator is linked to an unobserved Gaussian variable, z_t, by

$$S_t = 1\{z_t < 0\}, \tag{1}$$

where $1\{.\}$ is an indicator function that takes on a value of 1 if the condition in the brackets is met and 0 otherwise. Lower values of z_t therefore correspond to weaker business-cycle conditions and higher recession risk.

The VAR(p) is estimated for the augmented vector,

$$y_t = (x_t', z_t)', \tag{2}$$

that follows

$$y_t = c + B_1 y_{t-1} + \cdots + B_p y_{t-p} + u_t, \qquad u_t \sim N(0, \Sigma), \tag{3}$$

where u_t is the reduced-form residual.

Because one element of y_t is unobserved, the Qual-VAR treats the latent recession path and the VAR parameters as unknown objects to be estimated jointly, conditional on the observed binary recession dates. The posterior sampler alternates between two steps. First, conditional on a draw of the latent path $\{z_t\}_{t=1}^T$, we draw the VAR coefficients and a covariance matrix from the conjugate Bayesian VAR posterior. Following McCracken et al. (2022), each covariance draw is normalized by rescaling its last row and column so that the variance of the latent recession equation equals one. Second, conditional on the VAR parameters and the observed recession classification, each z_t is drawn from its normal conditional distribution, truncated to satisfy the sign restriction in Equation (1). Thus, z_t is drawn from the negative region in recession months and from the positive region in expansion months.

For each posterior draw, the forecast distribution of the latent factor determines the probability of recession through the event $z_{t+h} < 0$. Section 2.4 specifies how we evaluate this probability under the unconditional forecast and policy-intervention scenario.

2.2 Data

We estimate the Qual-VAR with six monthly lags on U.S. data from June 1976 to December 2019.[3] The estimation sample ends before the COVID-19 pandemic to prevent the extreme pandemic-period observations from shaping the reduced-form dynamics used in the counterfactual scenarios. The observed VAR vector is

$$x_t = (\Delta ip_t,\ \pi_t,\ r_t,\ R_t,\ a_t)', \tag{4}$$

where Δip_t is industrial production growth, π_t is annualized monthly CPI inflation, r_t is the federal funds rate, R_t is the 2-year Treasury yield, and a_t is the anticipated-policy indicator described below.[4]

We construct an anticipated-policy indicator from Blue Chip Economic Indicators forecasts of short-term interest rates [see Antolín-Díaz et al. (2021)]. The indicator summarizes revisions to the private sector expectation of the near-term policy path and is the observable variable used to identify shocks to anticipated monetary policy. Let $\widetilde{r}_{t|t+h}$ denote the Blue Chip forecast, formed at date t, of the short-term rate at period $t+h$. For each horizon $h = 0, 1, 2, 3$, the forecast revision is

$$\Delta\widetilde{r}_{t|t+h} = \widetilde{r}_{t|t+h} - \widetilde{r}_{t-1|t+h}.$$

Because revisions to expected policy rates across nearby horizons share a strong common component, Campbell et al. (2012) and Antolín-Díaz et al. (2021) motivate summarizing them with a low-dimensional measure. We therefore define the anticipated-policy indicator as the average revision over the current quarter and the next three quarters:

$$a_t = \frac{1}{4}\sum_{h=0}^{3} \Delta\widetilde{r}_{t|t+h}.$$

A positive value of a_t indicates an upward revision to expected short-term rates from the current quarter through the next three quarters. The indicator can therefore reflect both contemporaneous policy surprises and revisions to the expected future policy path. The structural restrictions in Section 2.3 separate these components when identifying the conventional monetary policy shock and anticipated monetary policy shock.

[3] Since the Blue Chip consensus forecast is available only from October 1980, we fill missing observations in the anticipated-policy indicator before estimation with a Kalman smoother, as described in Appendix A.1.

[4] Industrial production, the CPI, the federal funds rate, and the 2-year Treasury yield are obtained from FRED. Industrial production growth and CPI inflation are computed from log differences. The federal funds rate and the 2-year Treasury yield enter in percentage points.

The NBER recession indicator is available at the monthly frequency and enters the Qual-VAR through the sign of the latent recession factor described above. Because the NBER classifies business-cycle turning points rather than reporting a separate monthly recession variable, we construct the binary recession indicator S_t from the announced peak and trough dates. A peak month is coded as the last month of an expansion, $S_t = 0$, and the months after a peak through the subsequent trough are coded as recession months, $S_t = 1$. Accordingly, a trough month is the last month with $S_t = 1$.[5]

For episodes whose origin falls after December 2019, we keep the posterior distribution of the reduced-form parameters fixed at the December 2019 estimation sample and update the latent recession factor conditional on subsequently observed macro-financial variables through the month before the episode origin.[6] The NBER recession indicator does not enter this post-2019 latent-factor update. For example, for the July 2022 episode, we use observed macro-financial variables through June 2022 to update the latent recession factor before constructing the counterfactual scenario.

2.3 Structural Shock Identification

From the system of mutually orthogonal structural shocks, we jointly identify two monetary policy shocks for the scenario analysis. Joint identification within the same orthogonal structural system ensures that the two shocks represent distinct dimensions of monetary policy: a conventional monetary policy shock (MP) that captures an unexpected change in the current policy rate and an anticipated monetary policy shock (AMP) that captures news about the expected near-term path of the federal funds rate but leaves it unchanged on impact.

For each posterior draw of the reduced-form Qual-VAR, structural identification decomposes the reduced-form innovation as

$$u_t = A_0 \varepsilon_t, \qquad E(\varepsilon_t \varepsilon_t') = I, \qquad A_0 A_0' = \Sigma, \tag{5}$$

where u_t is the reduced-form innovation in Equation (3), ε_t is the vector of orthonormal structural shocks, and A_0 is the contemporaneous structural impact matrix. The implementation combines three types of identifying assumptions—sign, narrative, and exclusion

[5]The NBER recession indicator is used only through December 2019, the end of the Qual-VAR estimation sample. Because the sample ends before the February 2020 NBER peak, the end-of-sample treatment of potentially unannounced turning points discussed by Giusto and Piger (2017) and McCracken et al. (2022) is not required in the estimation.

[6]As a robustness check, we re-estimate the Qual-VAR through June 2022 using an adaptation of the pandemic-time-dummy approach of Cascaldi-Garcia (2022) and repeat the structural identification and scenario analysis. Appendix C.1 reports the results.

Table 1: Identification Restrictions for MP and AMP Shocks

Shock	Δip_0	π_0	z_0	r_0	a_0	r_h	Narrative
MP	$-$	$-$	$-$	$+$			1979:10
AMP	$-$	$-$	$-$	0	$+$	$+$	

Notes: The table reports the identifying restrictions for the conventional monetary policy shock (MP) and the anticipated monetary policy shock (AMP). The variables are industrial production growth (Δip_t), CPI inflation (π_t), the latent recession factor (z_t), the federal funds rate (r_t), and the anticipated-policy indicator (a_t). The entries $+$, $-$, and 0 denote positive, negative, and zero restrictions, respectively; empty cell indicates that no restriction is imposed. The column r_h refers to the federal funds rate response at horizons $h = 6, \ldots, 18$ months. The narrative restriction applies only to the MP shock and uses the October 1979 tightening episode.

restrictions—described here.

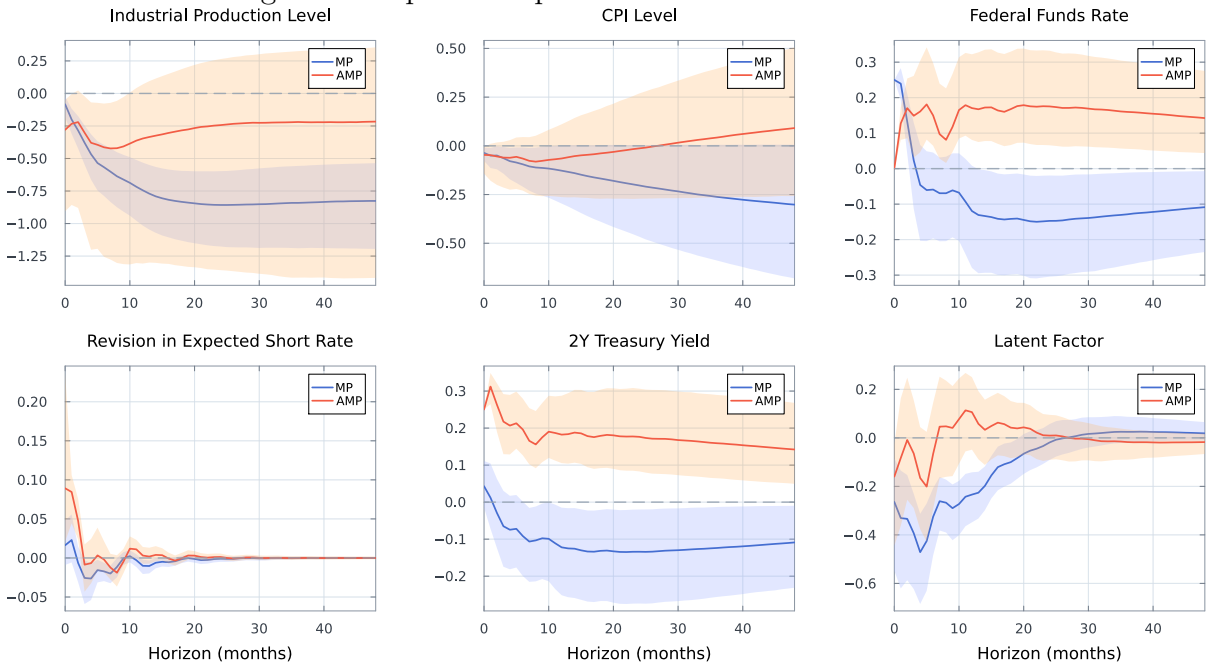
The sign restrictions follow the set-identification approach of Uhlig (2005): For each posterior draw, we retain rotations whose impulse responses satisfy the sign restrictions in Table 1. The narrative restriction follows Antolín-Díaz and Rubio-Ramírez (2018): It restricts the conventional monetary policy shock in a historical tightening episode (October 1979) using both its sign and its relative contribution to the policy-rate innovation. The impact zero restriction is imposed differently. It is built directly into the contemporaneous impact matrix, so that the AMP shock has no impact effect on the federal funds rate. Appendix A.3 gives the full algorithm.

Table 1 reports the identifying restrictions. The two shocks share the same impact restrictions on macroeconomic conditions: both lower industrial production growth, inflation, and the latent recession factor. Because z_t is positive in expansions and negative in recessions, a decline in z_t corresponds to higher recession risk. The shocks are distinguished by their policy-rate restrictions. A contractionary MP shock raises the federal funds rate on impact. A contractionary AMP shock has no contemporaneous effect on the funds rate but raises the anticipated-policy indicator on impact and is followed by a higher federal funds rate at medium horizons.

Figure 1 summarizes the impulse responses to the two identified policy shocks.[7] The MP shock is normalized to raise the federal funds rate by 25 basis points on impact, while the AMP shock is normalized to raise the 2-year Treasury yield by 25 basis points on impact. Both shocks are contractionary. Both shocks initially lower industrial production and the CPI price level. The conventional monetary policy shock produces a larger decline in real activity and in the latent recession factor, implying stronger recession-risk effects. The anticipated monetary policy shock loads more clearly on the expected-policy channel: It

[7]Appendix C.2 examines an additional zero impact response of the anticipated-policy indicator to the MP shock. The resulting impulse responses are similar to the benchmark specification.

Figure 1: Impulse Responses to MP and AMP Shocks



Notes: The figure reports posterior impulse responses to the conventional monetary policy shock (MP) and the anticipated monetary policy shock (AMP). Shaded regions denote 68% posterior uncertainty bands. MP shock is normalized to raise the federal funds rate by 25 basis points on impact. AMP shock is normalized to raise the 2-year Treasury yield by 25 basis points on impact.

raises revisions in expected short rates and the 2-year Treasury yield, while its effect on the latent factor is smaller.

2.4 Minimum-Effort Policy Scenarios

Given the identified shocks, we construct policy scenarios by choosing policy-shock paths from $h = 0, \ldots, H$ that achieve the inflation objective while limiting recession risk. We consider three policy specifications: (i) conventional monetary policy only (MP-only); (ii) anticipated monetary policy only (AMP-only); and (iii) both policy shocks (dual-policy). For each specification j, the available policy shocks at horizon h are

$$\boldsymbol{\varepsilon}_{j,t+h} = \begin{cases} \varepsilon_{\mathrm{MP},t+h}, & j = \mathrm{MP}, \\ \varepsilon_{\mathrm{AMP},t+h}, & j = \mathrm{AMP}, \\ (\varepsilon_{\mathrm{MP},t+h}, \varepsilon_{\mathrm{AMP},t+h})', & j = \mathrm{Dual}. \end{cases} \tag{6}$$

We refer to the MP-only and AMP-only specifications as single-policy specifications.

Let $\boldsymbol{\varepsilon}_{j,t,H}$ stack the available shocks over horizons $h = 0, \ldots, H$. The available shocks can

adjust over the entire forecast horizon, while all other structural shocks are set to zero.

Each scenario begins in month t, conditional on information through $t-1$, so horizon $h = 0$ denotes the first forecast month. For a given structural posterior draw, let $\widehat{\mathbf{y}}^u_{t,H}$ stack the unconditional mean forecasts from t through $t+H$, obtained by setting all future structural shocks to zero. Let $\widehat{\mathbf{y}}^c_{j,t,H}$ denote the corresponding scenario mean path and $\widehat{\boldsymbol{\varepsilon}}^c_{j,t,H}$ the policy-shock path that generates it. Iterating the identified VAR forward gives

$$\widehat{\mathbf{y}}^c_{j,t,H} = \widehat{\mathbf{y}}^u_{t,H} + M_{j,H}\widehat{\boldsymbol{\varepsilon}}^c_{j,t,H}, \tag{7}$$

where $M_{j,H}$ is the lower-block-triangular matrix of structural impulse responses to the policy shocks available under specification j.

The soft-landing scenario specifies an inflation target, Π^*; a target horizon, T_π; and a cap on model-implied recession probability, $\bar{p}$, over the forecast horizon $h = 0, \ldots, H$.[8] The inflation objective determines how much additional disinflation policy must deliver relative to the unconditional forecast. The target horizon specifies the time at which the central bank intends to arrive at the inflation objective. The cap limits the recession risk associated with the resulting path for all $h = 0, \ldots, H$.

We choose the policy-shock path that satisfies both objectives with the minimum sum of squared structural shocks. The resulting problem is

$$\begin{aligned} \widehat{\boldsymbol{\varepsilon}}^c_{j,t,H} = \arg\min_{\boldsymbol{\varepsilon}_{j,t,H}} \quad & \frac{1}{2}\boldsymbol{\varepsilon}'_{j,t,H}\boldsymbol{\varepsilon}_{j,t,H} \\ \text{subject to} \quad & \text{inflation restriction}, \\ & \text{recession-risk restriction}. \end{aligned} \tag{8}$$

Inflation objective. The inflation objective requires annualized monthly inflation to equal Π^* from horizon T_π through H. Let L_π select inflation at these horizons $h = T_\pi, \ldots, H$ from the stacked forecast vector. The restriction can be expressed in terms of the available policy shocks as

$$D_{\pi,j,H}\boldsymbol{\varepsilon}_{j,t,H} = \mathbf{f}_{\pi,t,H}, \tag{9}$$

[8] In the benchmark scenario, we set $\Pi^* = 2$, $T_\pi = 24$, $\bar{p} = 0.3$, and $H = 48$.

The calibrated value of $\bar{p}$ is broadly in line with recession-risk magnitudes in related studies. Rosenberg and Maurer (2008) use a 25 percent recession-signal threshold, while Ajello et al. (2022) note that recession probabilities of 25–35 percent are comparable to those estimated ahead of the 1994 monetary-policy tightening that resulted in a soft landing. In our 523-month sample spanning June 1976 to December 2019, 56 months are classified as recessions, corresponding to a historical recession frequency of approximately 10.7 percent. Appendix C.3 shows the results for alternative recession caps of 20 and 40 percent.

where

$$\begin{aligned} D_{\pi,j,H} &= L_\pi M_{j,H}, \\ \mathbf{f}_{\pi,t,H} &= \Pi^* \mathbf{1} - L_\pi \widehat{\mathbf{y}}^u_{t,H}. \end{aligned} \tag{10}$$

The matrix $D_{\pi,j,H}$ maps the available policy shocks into changes in inflation at the restricted horizons, while $\mathbf{f}_{\pi,t,H}$ gives the required adjustment relative to the unconditional inflation forecast at those horizons. $\mathbf{1}$ is an $(H - T_\pi + 1)$-dimensional vector of ones. Inflation at horizons before T_π is left unrestricted.

Recession-risk objective. The recession-risk objective requires the model-implied probability of recession to remain below $\bar{p}$ throughout the forecast horizon.

Let $\widehat{z}^c_{j,t+h}$ be the mean forecast of the latent recession factor at horizon h extracted from the stacked conditional forecast $\widehat{\mathbf{y}}^c_{j,t,H}$, and let $\widehat{\sigma}^{z,u}_{t+h}$ denote the unconditional forecast standard deviation of the latent factor at horizon h, held fixed for each posterior draw.[9] We evaluate recession probability as

$$p^c_{j,t+h} = \Phi\left(-\frac{\widehat{z}^c_{j,t+h}}{\widehat{\sigma}^{z,u}_{t+h}}\right). \tag{11}$$

The restriction $p^c_{j,t+h} \leq \bar{p}$ is equivalent to a linear lower bound on the scenario mean of the latent recession factor,

$$\widehat{z}^c_{j,t+h} \geq -\widehat{\sigma}^{z,u}_{t+h}\Phi^{-1}(\bar{p}), \qquad h = 0, \ldots, H. \tag{12}$$

Let L_z select the latent recession factor at these horizons, and let $\underline{\mathbf{z}}_{t,H}$ collect the corresponding lower bounds. The restriction can be expressed as

$$D_{z,j,H}\boldsymbol{\varepsilon}_{j,t,H} \geq \mathbf{f}_{z,t,H}, \tag{13}$$

where

$$\begin{aligned} D_{z,j,H} &= L_z M_{j,H}, \\ \mathbf{f}_{z,t,H} &= \underline{\mathbf{z}}_{t,H} - L_z \widehat{\mathbf{y}}^u_{t,H}. \end{aligned} \tag{14}$$

The matrix $D_{z,j,H}$ maps the available policy shocks into changes in the latent recession factor, while $\mathbf{f}_{z,t,H}$ gives the minimum adjustment required relative to its unconditional forecast.

Combining the inflation and recession-risk restrictions, we choose the policy-shock path that satisfies both objectives with the minimum sum of squared structural shocks. For each

[9] The scenario changes the expected path of the latent recession factor, but we hold forecast uncertainty fixed at its unconditional value for the same posterior draw. This keeps recession probabilities comparable across the unconditional and counterfactual paths.

policy specification j, the resulting problem is

$$\begin{aligned} \widehat{\boldsymbol{\varepsilon}}^{c}_{j,t,H} = \arg\min_{\boldsymbol{\varepsilon}_{j,t,H}} \quad & \frac{1}{2}\boldsymbol{\varepsilon}'_{j,t,H}\boldsymbol{\varepsilon}_{j,t,H} \\ \text{subject to} \quad & D_{\pi,j,H}\boldsymbol{\varepsilon}_{j,t,H} = \mathbf{f}_{\pi,t,H}, \\ & D_{z,j,H}\boldsymbol{\varepsilon}_{j,t,H} \geq \mathbf{f}_{z,t,H}. \end{aligned} \tag{15}$$

Relation to structural scenario analysis. Equation (15) builds on structural scenario analysis (Antolín-Díaz et al., 2021), which extends conventional conditional forecasting (Waggoner and Zha, 1999) by allowing restrictions on both future endogenous variables and identified structural shocks, and selects the shock path with the smallest sum of squared structural shocks. Two of our restrictions are of this kind: setting the non-policy shocks to zero restricts structural shocks, and the inflation objective restricts an endogenous variable. The recession-risk cap is not, because structural scenario analysis imposes equality restrictions only. Imposing the cap as an equality at every horizon would hold recession probability at exactly $\bar{p}$ in every month, forcing policy to raise recession risk where the unconditional outlook is already below the cap. With only one policy shock, it would also impose more restrictions than there are free shock entries. Imposing the cap only where it binds would pose the right problem, but those horizons depend on the shock path and are not known in advance. The inequality formulation resolves this by imposing the cap directly, so that the horizons at which it binds are determined jointly with the shock path. Appendix A.4 sets out structural scenario analysis and counts the restrictions under both formulations.

Loss function and feasibility. The objective weights every shock entry equally, because the structural shocks are mutually independent with unit variance. Under normality of the structural shocks, it equals the negative log likelihood of the policy-shock path up to an additive constant. The solution is therefore the most likely policy-shock path that delivers both objectives. Because $\widehat{\sigma}^{z,u}_{t+h}$ is held fixed, every constraint is linear in the shock path, and Equation (15) is a convex quadratic program with a unique solution whenever its constraints can be jointly satisfied.

The constraints can fail to be jointly satisfied when, for a given posterior draw, no shock path that delivers the inflation objective keeps recession probability at or below the cap at every horizon. Although the inflation equality constraints can be satisfied using either policy channel, the resulting shock paths may not be able to keep recession probability below the specified cap at every horizon. Infeasibility therefore arises when no combination of available policy shocks simultaneously satisfies both objectives.[10]

[10]Unlike the equality-based structural scenario analysis, the inequality-constrained formulation does not

When the constraints cannot be jointly satisfied for a posterior draw, we classify the scenario as infeasible and do not construct a policy-shock path for that draw. Infeasible draws remain in the denominator when calculating the posterior feasibility share. Policy-path summary measures are computed only for draws with feasible solutions.

2.5 Scenario Summary Measures

Let $d = 1, \ldots, N$ index structural posterior draws and $j \in \{\text{MP}, \text{AMP}, \text{Dual}\}$ index the three policy specifications. For each scenario, we compute summary measures from the forecast mean paths and fitted structural shocks using the corresponding structural parameters.

Policy effort. Policy effort is the root-mean-square magnitude of the available policy shocks over the forecast horizon $h = 0, \ldots, H$,

$$E_{j,t}^{(d)} = \left[\frac{1}{H+1} \sum_{h=0}^{H} \left\| \widehat{\boldsymbol{\varepsilon}}_{j,t+h}^{(d)} \right\|^2 \right]^{1/2}. \tag{16}$$

Peak recession probability. For each draw, peak recession probability is

$$P_{j,t}^{(d)} = \max_{0 \le h \le H} p_{j,t+h}^{(d)}. \tag{17}$$

The benchmark uses $H = 48$, with recession probabilities evaluated as in Equation (11).

Feasibility share. A risk-constrained scenario is feasible for a posterior draw if there exists a policy-shock path that satisfies both the inflation and recession-risk restrictions in Equation (15). The feasibility share is

$$\text{FS}_{j,t} = \frac{1}{N} \sum_{d=1}^{N} \mathbf{1} \left\{ \text{both objectives are feasible for draw } d \right\}. \tag{18}$$

This statistic measures the posterior share of draws for which the available policy shocks can satisfy both objectives simultaneously.

Policy contributions and the AMP share. Let $e'_{\pi,h}$ select monthly inflation at horizon h from the stacked forecast vector. Let $\widehat{\boldsymbol{\varepsilon}}_{k,t,H}^{(d)}$ collect the fitted entries of the dual-policy

require the total number of recession-risk restrictions to be smaller than the number of free shock entries. Instead, feasibility requires the existence of a policy-shock path that jointly satisfies the inflation equalities and all recession-risk inequalities.

shock path for channel k, and let $M_{k,H}^{(d)}$ be the corresponding impulse-response matrix in Equation (7). The signed contribution of channel k to disinflation at the target horizon is

$$\delta_{k,t}^{(d)}(T_\pi) = -e'_{\pi,T_\pi} M_{k,H}^{(d)} \widehat{\boldsymbol{\varepsilon}}_{k,t,H}^{(d)}, \qquad k \in \{\text{MP}, \text{AMP}\}. \tag{19}$$

At the target horizon, the required policy disinflation equals the sum of the contributions from the two policy channels.

The AMP share is defined using the magnitudes of these contributions,

$$\omega_{\text{AMP},t}^{(d)} = \frac{\left|\delta_{\text{AMP},t}^{(d)}(T_\pi)\right|}{\left|\delta_{\text{MP},t}^{(d)}(T_\pi)\right| + \left|\delta_{\text{AMP},t}^{(d)}(T_\pi)\right|}. \tag{20}$$

3 Soft-Landing Scenario Results

This section evaluates the soft-landing counterfactuals in two steps. We first use May 1981 and July 2022 as worked examples to explain the policy-design mechanism. We then turn to the 24 representative starting dates to assess how these relationships vary across a broader range of historical conditions.

3.1 Illustrative Historical Episodes

May 1981 and July 2022 serve as worked examples of the inverse policy-design problem.[11] Table 2 compares the two episodes in three steps. Panel A summarizes the initial inflation and pre-intervention policy burden. Panel B reports the policy path that meets the inflation objective alone with the smallest overall shock movements. Panel C shows how imposing the recession-risk cap changes policy effort and the MP–AMP composition.

Panel A shows why observed inflation is an incomplete guide to the policy challenges. CPI inflation over the preceding 12 months was 9.7 percent in May 1981 and 8.6 percent in July 2022. However, required policy disinflation at month 24 is 4.8 percentage points in May 1981, more than twice the 2.2 percentage points required in July 2022. The unconditional recession outlook also differs sharply. Peak recession probability over months 0–48 is 81.7 percent in May 1981 but only 22.6 percent in July 2022. Therefore, May 1981 combines a larger remaining disinflation burden with a recession outlook already above the cap, while July 2022 begins with a smaller disinflation burden and an unconditional peak recession probability below the cap.

[11] Appendix B.1 reports the corresponding benchmark counterfactuals for six historically salient U.S. anti-inflation episodes.

Table 2: Policy Design in May 1981 and July 2022

	May 1981	July 2022
A. Initial Conditions and Policy Burden		
Initial 12-month CPI inflation (%)	9.7	8.6
Required policy disinflation at month 24 (pp)	4.8	2.2
Peak recession probability, unconditional forecast (%)	81.7	22.6
B. Inflation-only solution		
Policy effort (RMS, s.d.)	0.90	0.40
AMP share of policy-induced disinflation (%)	30.0	31.6
Peak recession probability (%)	98.0	48.3
C. Risk-Constrained Solution		
Policy effort (RMS, s.d.)	2.09	0.52
AMP share of policy-induced disinflation (%)	75.3	46.6

Notes: Required policy disinflation at month 24 is the reduction in annualized one-month CPI inflation required at month 24, relative to the unconditional forecast, to reach the 2 percent target. Peak recession probability is the maximum over months $0-48$ under the path reported in Panels A and B. Panel B reports the minimum-effort combination of MP and AMP shocks that satisfies the inflation objective without imposing the recession-risk cap. Panel C adds the 30 percent cap. Policy effort is the root-mean-square magnitude of available policy shocks over 0–48 months. The AMP share is the percentage of policy-induced disinflation attributable to AMP shocks. Entries are posterior medians.

Panel B shows what happens when policy shocks are chosen solely to satisfy the inflation objective. The minimum-effort intervention requires 0.90 standard-deviation units of policy effort in May 1981 and 0.40 in July 2022. Despite this substantial difference in the magnitude of the intervention, its composition is nearly identical across the two episodes.[12] AMP shocks account for 30.0 percent of policy-induced disinflation in May 1981 and 31.6 percent in July 2022. Under the inflation-only solution, peak recession probability rises from 81.7 to 98.0 percent in May 1981 and from 22.6 to 48.3 percent in July 2022. The July 2022 episode is particularly informative, as recession risk begins below the 30 percent cap, but the policy intervention needed to deliver the inflation objective pushes peak recession probability above it.

Panel C shows how explicitly constraining recession risk changes the policy intervention, particularly its composition. The impulse responses in Section 2.3 indicate that MP shocks

[12]This similarity reflects the structure of the problem. Under the inflation objective alone, the minimum-effort shock path is linear in the required disinflation, so a larger requirement scales up both MP and AMP shocks without changing their mix. Because the impulse responses are the same at every starting date, differences in the amount of disinflation required change the size of the intervention more than its composition.

deliver stronger disinflation but also raise recession risk more than AMP shocks. Thus, once recession risk is constrained, the policy mix shifts toward less recessionary AMP shocks. Because AMP shocks have a weaker disinflationary effect, larger shock movements are needed to deliver the required inflation reduction. Therefore, policy effort and the AMP share rise in both episodes. In May 1981, policy effort rises from 0.90 to 2.09 standard-deviation units, and the AMP share rises from 30.0 to 75.3 percent. In July 2022, policy effort rises from 0.40 to 0.52, while the AMP share increases from 31.6 to 46.6 percent. The adjustment is especially large in May 1981, when a substantial remaining disinflation burden must be delivered against an already adverse recession outlook.

3.2 Representative Starting Conditions

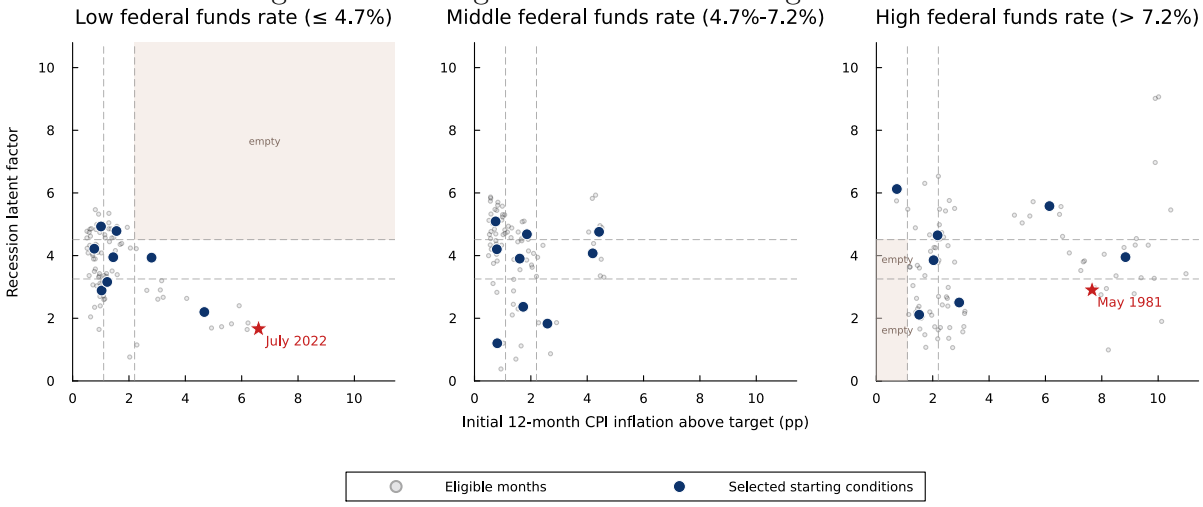
We extend the analysis to 24 representative historical starting conditions to assess how the relationships illustrated by May 1981 and July 2022 vary across the observed economic environment. We begin with the 284 months from June 1977 through July 2022 in which the economy was not classified as being in an NBER recession in the preceding month and 12-month CPI inflation exceeded the 2 percent target by more than 0.5 percentage point.

Each eligible month is characterized by three conditions: initial inflation pressure, model-implied latent recession factor, and the federal funds rate. For each condition, we classify the month as low, middle, or high across eligible months. Combining the three classifications produces $3 \times 3 \times 3 = 27$ possible types of starting conditions. For example, one type combines high inflation pressure, low latent recession factor, and a high federal funds rate. Every eligible month belongs to one of these 27 types.

24 of the 27 possible combinations occur in the historical data. From each observed type, we select the actual historical month closest to the within-type medians of the three standardized classification variables. The resulting set spans the observed range of the three dimensions without weighting combinations by their historical frequency. Appendix B.2 reports the numerical cutoffs, the number of eligible months in each type, and the selected historical dates.

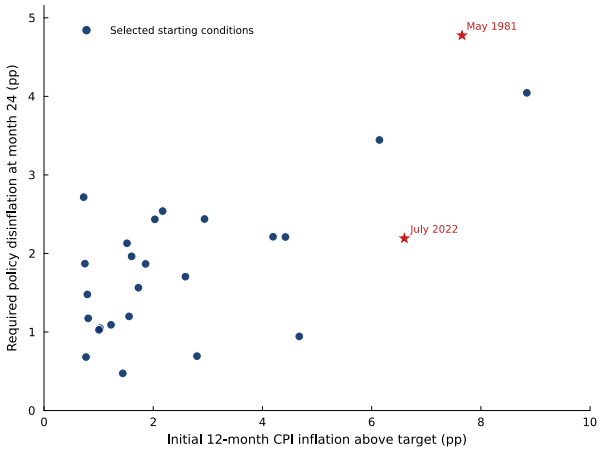
Figure 2 illustrates the resulting coverage. Relative to the two illustrative episodes in Section 3.1, the selected starting conditions extend the analysis toward lower-inflation environments. At the same time, the selection retains potentially demanding starting points, including combinations of high inflation, low latent recession factor, and high policy rates.

Figure 2: Coverage of Historical Starting Conditions



Notes: Gray dots show the 284 eligible starting months from June 1977 to July 2022. A month t is eligible when month $t-1$ is not classified as an NBER recession and 12-month CPI inflation at $t-1$ exceeds the 2 percent target by more than 0.5 percentage points. The horizontal axis shows initial 12-month CPI inflation above target. The vertical axis reports the posterior median of the model-implied recession risk factor from Qual-VAR; lower values indicate weaker business-cycle conditions and greater recession vulnerability. Panels classify the federal funds rate using the 33rd- and 66th-percentile cutoffs of the eligible-month distribution. Dashed lines apply the same classification to inflation pressure and the model-implied recession risk factor. Every eligible month belongs to one of 27 possible combinations of low, middle, or high conditions. Filled circles denote the historical month selected as most typical of each of the 24 populated combinations. Stars denote May 1981 and July 2022 episodes.

Figure 3: Initial Inflation and Required Policy Disinflation



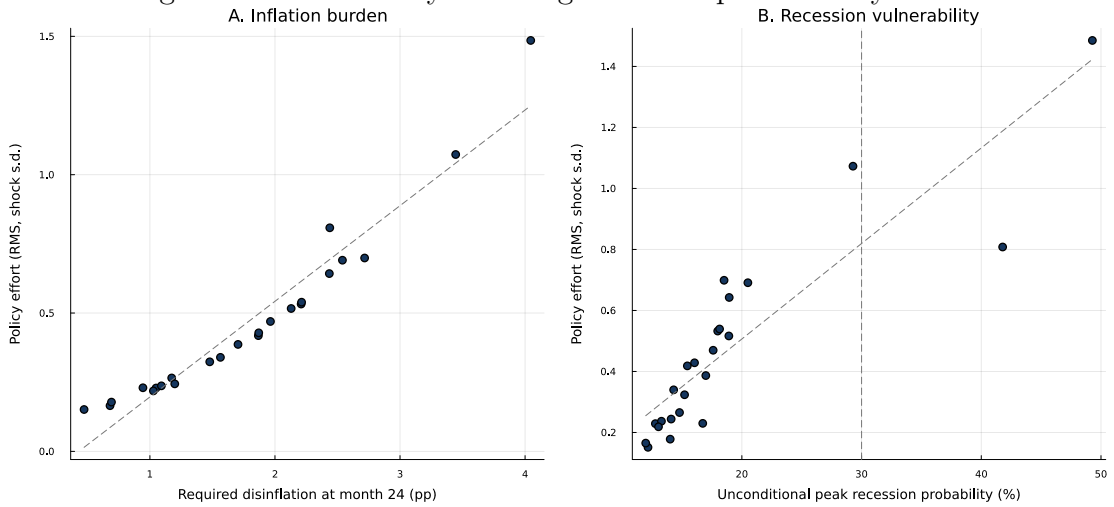
Notes: Circles denote the 24 selected starting conditions, and stars denote May 1981 and July 2022. The horizontal axis reports 12-month CPI inflation at $t-1$ minus 2 percent target. The vertical axis reports required policy disinflation at month 24, defined as the additional reduction in annualized monthly CPI inflation needed to bring the unconditional forecast to 2 percent. The plotted points are posterior medians.

3.2.1 From Initial Conditions to the Policy Challenge

Figure 3 relates initial 12-month CPI inflation above target to required policy disinflation at month 24. Initial 12-month inflation characterizes the inflation environment, while required policy disinflation measures the additional reduction in annualized monthly CPI inflation that policy shocks must deliver at the target horizon after accounting for the unconditional forecast.

The figure exhibits a broad positive relationship, with higher initial inflation generally requiring more disinflation from policy. However, the relationship is far from one-for-one. Similar initial 12-month inflation can correspond to substantially different policy burdens because the amount of disinflation already embedded in the unconditional forecast varies across starting dates. Conversely, some starting dates with moderate initial inflation still require substantial policy intervention because inflation remains persistent in the unconditional forecast. Thus, as illustrated by the comparison between May 1981 and July 2022, initial inflation is informative about the starting environment, but it does not by itself rank the difficulty of bringing inflation down to target.

Figure 4: Initial Policy Challenges and Required Policy Effort



Notes: Each point represents one of the 24 selected starting conditions. Panel A relates required policy disinflation at month 24 to policy effort, while Panel B reports unconditional peak recession probability to policy effort. Required policy disinflation is the additional reduction in annualized monthly CPI inflation needed at month 24 to reach the 2 percent target under the unconditional forecast. Unconditional peak recession probability is the posterior median of the maximum under the unconditional forecast over months 0–48. Policy effort is measured under the risk-constrained dual-policy scenario, in which MP and AMP shocks adjust to satisfy the inflation objective and the 30 percent recession-risk cap. Policy effort is the root-mean-square magnitude of available policy shocks over 0–48 months. The vertical reference line in the right panel marks the 30 percent cap. Dashed lines are OLS fits across the 24 selected conditions.

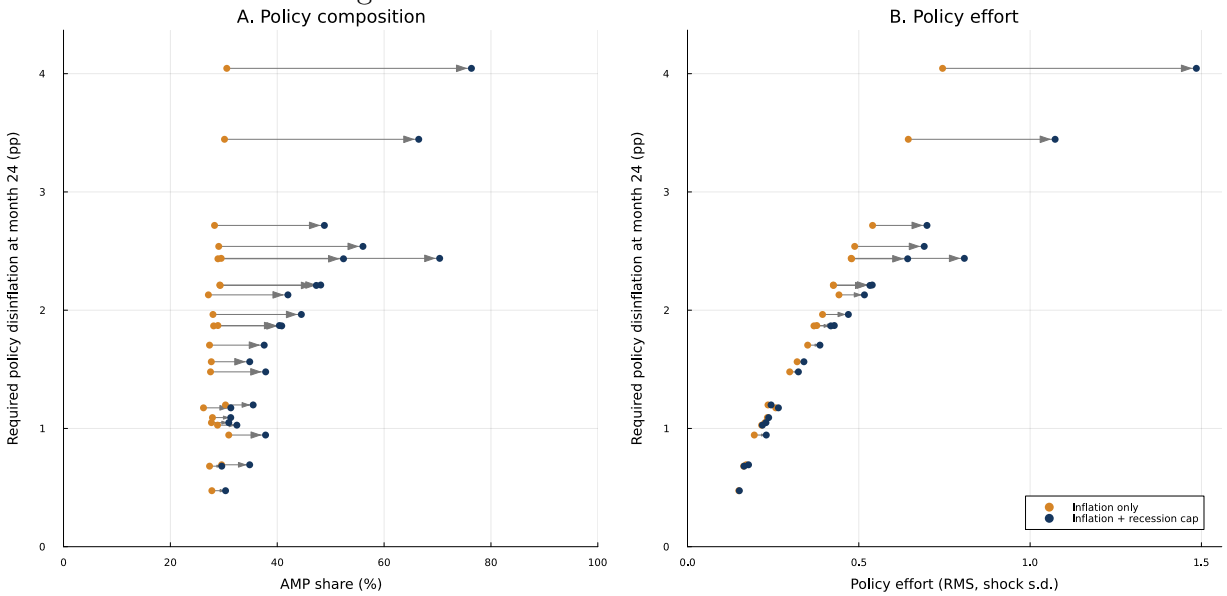
Figure 4 links the two pre-intervention policy challenges to the policy effort required under the risk-constrained scenario. Panel A reports required policy disinflation at month

24, while Panel B reports peak recession probability under the unconditional forecast. Both panels report the minimum policy effort needed to satisfy the inflation objective and the 30 percent recession-risk cap when MP and AMP shocks are jointly available.

Policy effort is positively associated with both measures. Panel A shows that larger required disinflation is associated with greater policy effort. Panel B shows a similar positive relationship between unconditional peak recession probability and policy effort. However, these relationships are not independent. Across the selected starting conditions, required policy disinflation and unconditional peak recession probability are strongly correlated (0.78). Starting conditions with a larger inflation pressure often also tend to have a less favorable recession outlook.

3.2.2 The Cost of Limiting Recession Risk

Figure 5: Cost of Recession-Risk Control



Notes: The figure compares inflation-only and risk-constrained dual-policy scenarios across 24 selected starting conditions. Panel A reports the AMP share of policy-induced disinflation, while Panel B reports policy effort. The vertical axis shows required policy disinflation at month 24 in both panels. Orange circles denote the inflation-only dual-policy scenario, while blue circles denote the risk-constrained dual-policy scenario that additionally satisfies the 30 percent recession-risk cap. Horizontal lines connect the two solutions for each starting condition.

Figure 5 shows how imposing the recession-risk cap changes policy composition and required policy effort. Both scenarios satisfy the same inflation objective, but only the risk-constrained scenario limits recession probability.

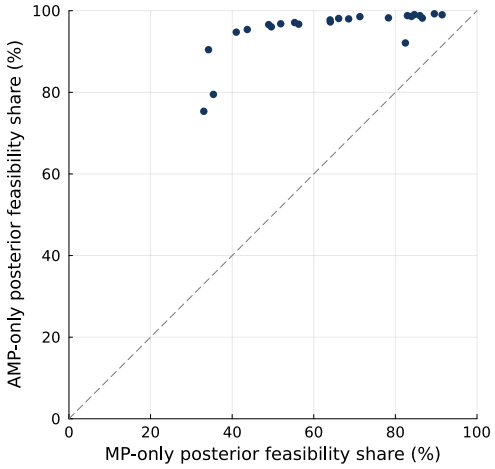
Panel A compares the AMP share of policy-induced disinflation under the two scenarios.

The inflation-only scenario relies more heavily on the MP shocks, which deliver greater disinflation per unit of shock. Once recession risk is constrained, AMP becomes more attractive because its recessionary effects are smaller. The AMP share generally increases, with larger changes occurring when more policy disinflation is required. This pattern also reflects differences in initial conditions. Starting dates with greater required disinflation tend to have higher unconditional recession risk, making the recession-risk constraint more consequential for the policy mix.

Panel B shows that imposing the recession-risk cap also increases the required policy effort. The largest increases occur when more policy disinflation is required. Greater reliance on AMP helps limit recession risk, but its weaker disinflationary effect needs larger overall shock movements.

3.2.3 Single-Policy Feasibility

Figure 6: Feasibility of Single-Policy Interventions



Notes: The figure compares posterior feasibility shares under MP-only and AMP-only scenarios across the 24 selected starting conditions. Feasibility share is the proportion of posterior draws for which both the inflation objective and the 30 percent recession-risk cap can be satisfied. The dashed diagonal line denotes the 45-degree line.

Figure 6 compares the posterior feasibility shares of MP-only and AMP-only interventions across the 24 selected starting conditions. Feasibility is defined as the proportion of posterior draws for which the available policy channel can satisfy both the inflation objective and the 30 percent recession-risk cap.

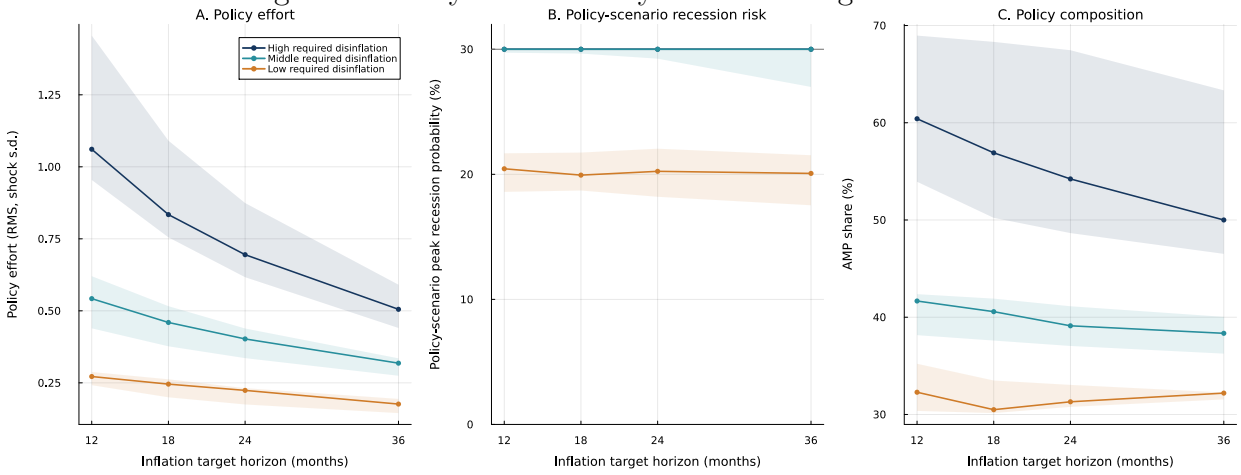
All points lie above the 45-degree line, indicating that AMP-only interventions are more likely to satisfy both objectives than MP-only interventions. Under AMP-only, feasibility shares range from 79.5 to 99.3 percent, compared with 34.2 to 91.4 percent under MP-

only. This difference is consistent with the weaker recessionary effects of AMP shocks. The greater feasibility of AMP-only interventions helps explain why imposing the recession-risk constraint shifts the dual-policy intervention toward AMP, as shown in Figure 5.

3.2.4 The Role of the Inflation-Target Horizon

We next ask how the policy problem changes when policymakers have more time to reach the inflation target. Figure 7 varies the inflation-target horizon, T_π, from 12 to 36 months while holding the recession-risk cap fixed at 30 percent. We divide the 24 selected starting conditions into high-, middle-, and low-required-disinflation groups based on required policy disinflation at the benchmark 24-month horizon.

Figure 7: Policy Outcomes by Inflation-Target Horizon



Notes: The figure reports policy outcomes across the 24 selected starting conditions as the inflation-target horizon, T_π, varies from 12 to 36 months, with the recession-risk cap fixed at 30 percent and policy shocks allowed to adjust over months 0–48. Panel A reports policy effort, measured as the root-mean-square magnitude of the available policy shocks over months 0–48. Panel B reports peak recession probability under the risk-constrained policy scenario. Panel C reports the AMP share of policy-induced disinflation. The selected conditions are divided into three fixed groups of eight according to required policy disinflation at the benchmark 24-month horizon. Lines denote within-group medians, and shaded bands denote interquartile ranges.

Panel A shows that a longer target horizon reduces policy effort in all three groups. The decline is largest in the high-required-disinflation group, where policy effort is highest at shorter horizons. The middle- and low-required-disinflation groups show smaller declines because policy effort is already comparatively modest. Despite this reduction in policy effort, Panel B shows that peak recession probability changes little. In the high- and middle-disinflation groups, it stays at the 30 percent cap at every target horizon. Because the cap is a ceiling, this means the constraint continues to bind: even with 36 months to reach the target, the minimum-effort path still raises recession risk to the cap. A lower recession

probability is attainable, but only with additional policy effort. In the low-disinflation group, where the cap does not bind, peak recession probability stays near 20 percent.

Panel C shows how the minimum-effort policy uses the additional time. MP shocks generate more disinflation per unit of shock but also have stronger recessionary effects than AMP shocks. When more disinflation is required within a short horizon, the binding cap limits reliance on MP, leading to greater use of AMP. As the target horizon lengthens, the smaller intervention leaves room to rely more on MP while keeping recession risk at the cap. Consequently, the AMP share declines in the high- and middle-disinflation groups: the additional time is spent on a smaller and more conventional intervention rather than on lowering recession risk. In the low-required-disinflation group, where the cap does not bind, the AMP share changes little.

Table 3: Required Policy Disinflation by Target Horizon

	12 months	18 months	24 months	36 months
High required disinflation	2.92	2.73	2.56	2.29
Middle required disinflation	1.64	1.71	1.75	1.73
Low required disinflation	0.81	0.88	0.91	0.94

Notes: The table reports median required policy disinflation, in percentage points, across three fixed groups of eight selected starting conditions. The groups are defined by required policy disinflation at the benchmark 24-month horizon and are held fixed as the target horizon varies. Required policy disinflation is the additional reduction in annualized monthly CPI inflation needed to bring the unconditional forecast to the 2 percent target at the indicated horizon.

Table 3 provides further insight into these differences. In the high-required-disinflation group, extending the target horizon from 12 to 36 months reduces required policy disinflation from 2.92 to 2.29 percentage points. The unconditional inflation forecast moves closer to target over the additional period, leaving less disinflation for policy to deliver. In contrast, required policy disinflation changes little in the middle and low groups, rising slightly from 1.64 to 1.73 percentage points and from 0.81 to 0.94 percentage points, respectively.

In the low-required-disinflation group, where the recession-risk cap is less restrictive, both required policy disinflation and the policy mix change little as the target horizon increases. Giving policymakers more time allows them to achieve a similar inflation reduction with smaller shocks spread over a longer period. However, because the effects of monetary policy on economic activity are persistent, these smaller shocks can still generate recessionary pressure as their effects accumulate over time. As a result, a longer horizon may delay the peak in recession risk without substantially reducing its magnitude.

4 Conclusion

References

Ajello, Andrea, Luca Benzoni, Makena Schwinn, Yannick Timmer, and Francisco Vazquez-Grande, "Monetary policy, inflation outlook, and recession probabilities," *Economic Perspectives*, 2022, (4).

Antolín-Díaz, Juan and Juan F Rubio-Ramírez, "Narrative sign restrictions for SVARs," *American Economic Review*, 2018, *108* (10), 2802–2829.

_ , Ivan Petrella, and Juan F Rubio-Ramirez, "Structural scenario analysis with SVARs," *Journal of Monetary Economics*, 2021, *117*, 798–815.

Ball, Laurence, "What determines the sacrifice ratio?," in "Monetary policy," The University of Chicago Press, 1994, pp. 155–193.

Barnichon, Régis and Geert Mesters, "A sufficient statistics approach for macro policy," *American Economic Review*, 2023, *113* (11), 2809–2845.

Blinder, Alan S, "Landings, soft and hard: The federal reserve, 1965–2022," *Journal of Economic Perspectives*, 2023, *37* (1), 101–120.

Boissay, Frederic, Fiorella De Fiore, and Enisse Kharroubi, "Hard or soft landing?," BIS Bulletins 59, Bank for International Settlements 2022.

Bullard, James, Alex Grimaud, Isabelle Salle, and Gauthier Vermandel, "Soft Landing and Inflation Scares," *Journal of Monetary Economics*, 2026, *157*, 103871.

Campbell, Jeffrey R, Charles L Evans, Jonas DM Fisher, Alejandro Justiniano, Charles W Calomiris, and Michael Woodford, "Macroeconomic effects of federal reserve forward guidance [with comments and discussion]," *Brookings papers on economic activity*, 2012, pp. 1–80.

Cascaldi-Garcia, Danilo, "Pandemic priors," *International Finance Discussion Paper*, 2022, (1352).

Cecchetti, Stephen G and Robert W Rich, "Structural estimates of the US sacrifice ratio," *Journal of Business & Economic Statistics*, 2001, *19* (4), 416–427.

Cecchetti, Stephen, Michael Feroli, Peter Hooper, Frederic S Mishkin, and Kermit L Schoenholtz, "Managing disinflations," Technical Report, CEPR Discussion Papers 2023.

de Soyres, François and Zina Saijid, "Lessons from Past Monetary Easing Cycles," May 2024. FEDS Notes.

Domash, Alex and Lawrence H Summers, "A labor market view on the risks of a US hard landing," *Journal of Policy Modeling*, 2022, *44* (4), 758–767.

Dueker, Michael, "Dynamic forecasts of qualitative variables: a Qual VAR model of US recessions," *Journal of Business & Economic Statistics*, 2005, *23* (1), 96–104.

Erceg, Christopher J and Andrew T Levin, "Imperfect credibility and inflation persistence," *Journal of monetary economics*, 2003, *50* (4), 915–944.

Fischer, Stanley, "Contracts, credibility, and disinflation," Technical Report, National Bureau of Economic Research 1984.

Giannone, Domenico, Michele Lenza, and Giorgio E Primiceri, "Prior selection for vector autoregressions," *Review of Economics and Statistics*, 2015, *97* (2), 436–451.

Giusto, Andrea and Jeremy Piger, "Identifying business cycle turning points in real time with vector quantization," *International Journal of Forecasting*, 2017, *33* (1), 174–184.

Gordon, Robert J and Stephen R King, "The output cost of disinflation in traditional and vector autoregressive models," *Brookings Papers on Economic Activity*, 1982, *1982* (1), 205–244.

Hakamada, Mai and Carl E. Walsh, "The Consequences of Falling Behind the Curve: Inflation Shocks and Policy Delays under Rational and Behavioral Expectations," IMF Working Paper 2024/042, International Monetary Fund March 2024.

McCracken, Michael W, Joseph T McGillicuddy, and Michael T Owyang, "Binary conditional forecasts," *Journal of Business & Economic Statistics*, 2022, *40* (3), 1246–1258.

McKay, Alisdair and Christian K Wolf, "What can time-series regressions tell us about policy counterfactuals?," *Econometrica*, 2023, *91* (5), 1695–1725.

Okun, Arthur M, "Efficient disinflationary policies," *The American Economic Review*, 1978, *68* (2), 348–352.

Pariès, Darracq Matthieu, Antoine Kornprobst, and Romanos Priftis, "Monetary policy strategies to navigate post-pandemic inflation: An assessment using the ECB's New Area-Wide Model," Technical Report, ECB Working Paper 2024.

Penrose, Roger, "On best approximate solutions of linear matrix equations," in "Mathematical Proceedings of the Cambridge Philosophical Society," Vol. 52 Cambridge University Press 1956, pp. 17–19.

Romer, Christina D and David H Romer, "Does monetary policy matter? A new test in the spirit of Friedman and Schwartz," *NBER macroeconomics annual*, 1989, *4*, 121–170.

_ **and** _ , "Monetary policy matters," *Journal of Monetary Economics*, 1994, *34* (1), 75–88.

_ **and** _ , "Presidential address: Does monetary policy matter? The narrative approach after 35 years," *American Economic Review*, 2023, *113* (6), 1395–1423.

Rosenberg, Joshua V and Samuel Maurer, "Signal or noise? Implications of the term premium for recession forecasting," *Federal Reserve Bank of New York-Economic Policy Review*, 2008, *14* (1).

Stellato, Bartolomeo, Goran Banjac, Paul Goulart, Alberto Bemporad, and Stephen Boyd, "OSQP: An Operator Splitting Solver for Quadratic Programs," *Mathematical Programming Computation*, 2020, *12* (4), 637–672.

Svensson, Lars EO, "Inflation forecast targeting: Implementing and monitoring inflation targets," *European economic review*, 1997, *41* (6), 1111–1146.

_ **and Michael Woodford**, "Implementing optimal policy through inflation-forecast targeting," in "The inflation-targeting debate," University of Chicago Press, 2004, pp. 19–92.

Uhlig, Harald, "What are the effects of monetary policy on output? Results from an agnostic identification procedure," *Journal of Monetary Economics*, 2005, *52* (2), 381–419.

Verbrugge, Randal and Saeed Zaman, "The hard road to a soft landing: Evidence from a (modestly) nonlinear structural model," *Energy Economics*, 2023, *123*, 106733.

Waggoner, Daniel F and Tao Zha, "Conditional forecasts in dynamic multivariate models," *Review of Economics and Statistics*, 1999, *81* (4), 639–651.

Woodford, Michael, "The case for forecast targeting as a monetary policy strategy," *Journal of economic perspectives*, 2007, *21* (4), 3–24.

A Computational Details

This appendix collects the implementation details behind Section 2, following the order of the empirical pipeline: the data transformations and missing-data treatment, the Qual-VAR sampler, the structural-identification step, and the minimum-effort scenario solution used to construct the soft-landing counterfactuals.

A.1 Data Transformation and Missing-Data Imputation

Let X_t denote the raw monthly data vector before transformation. Industrial production enters as log differences multiplied by 100, while the CPI enters as log differences multiplied by 1200. The federal funds rate, the 2-year Treasury yield, and the anticipated-policy indicator enter in levels. The NBER recession indicator enters as the observed qualitative variable and is placed last in the estimation matrix, an implementation convention because the Qual-VAR sampler treats the final column as the qualitative variable.

The observed recession indicator is recoded before estimation. A recession month is coded as -1 and an expansion month as $+1$. Thus, we recoded -1 with 0 for S_t. The latent Gaussian recession variable is then sampled so that recession observations correspond to the negative latent region. This coding implies

$$S_t = 1\{z_t < 0\}, \tag{A.1}$$

where $S_t = 1$ denotes an NBER recession month. A negative response of z_t means a movement toward recession.

Some observable variables have missing values. We fill these observations before estimating the Qual-VAR. The imputation procedure uses only the continuous observable variables; it does not impute the qualitative recession indicator. Let Y_t denote the transformed observable vector excluding the qualitative variable. The imputation proceeds in four steps.

First, we find the longest contiguous block of observations with no missing entries. This block supplies the regression sample for a preliminary VAR with the same lag length used in the VAR. Second, we estimate a Minnesota-prior BVAR on this complete block. The prior shrinks higher-order lags more heavily and uses variable-specific innovation scales based on univariate autoregressions. Third, we write the fitted BVAR in companion form and run a Kalman filter and Rauch–Tung–Striebel smoother on the full sample. The filter omits missing entries from the measurement update in each month. Fourth, we replace each originally missing cell with its smoothed value. Observed entries are left unchanged.

A.2 Reduced-Form Qual-VAR Sampler

This subsection describes the posterior sampler for the reduced-form Qual-VAR. The sampler treats the latent recession factor as missing data. Conditional on a path for the latent factor, the model is a standard Bayesian VAR. Conditional on the VAR parameters, the latent factor is drawn from Gaussian distributions that are truncated to match the observed NBER recession classification. The sampler alternates between these two blocks.

Let x_t denote the vector of observed continuous variables, and let z_t denote the latent recession factor. The augmented vector is

$$y_t = (x_t', z_t)'. \tag{A.2}$$

The observed recession indicator is

$$S_t = 1\{z_t < 0\}, \tag{A.3}$$

where $S_t = 1$ in NBER recession months and $S_t = 0$ otherwise. This convention implies that recessions correspond to negative values of z_t, while expansions correspond to positive values.

The augmented reduced-form VAR is

$$y_t = c + B_1 y_{t-1} + \cdots + B_p y_{t-p} + u_t, \qquad u_t \sim N(0, \Sigma). \tag{A.4}$$

Let

$$X_t = (1, y_{t-1}', \ldots, y_{t-p}')', \tag{A.5}$$

and collect the intercept and lag coefficients in the coefficient matrix B, where

$$B' = \begin{bmatrix} c & B_1 & \cdots & B_p \end{bmatrix}. \tag{A.6}$$

Then the VAR can be written as

$$y_t = B' X_t + u_t. \tag{A.7}$$

Stacking the observations after the initial lags gives

$$Y = XB + U, \qquad \mathrm{vec}(U) \sim N(0, \Sigma \otimes I_T), \tag{A.8}$$

where Y is the $T \times n$ matrix of augmented observations, X is the $T \times k$ matrix of regressors, $k = 1 + np$, T is the number of usable observations after the initial lags, n is the number of

variables in y_t, and $\otimes$ denotes the Kronecker product.

A.2.1 First block: drawing the VAR parameters

The first block draws the reduced-form coefficients and covariance matrix conditional on the current latent path. We use a conjugate Minnesota-style Normal-inverse-Wishart prior, following the GLP prior-selection logic in Giannone et al. (2015) and the Qual-VAR implementation in McCracken et al. (2022). Let B_0 denote the prior mean of the full coefficient matrix B. The first row of B_0 is the prior mean for the intercept c, and the remaining rows are the prior centers for the lag coefficients. The prior is

$$\operatorname{vec}(B) \mid \Sigma, \lambda \sim N\left(\operatorname{vec}(B_0),\, \Sigma \otimes V_\lambda\right), \tag{A.9}$$

$$\Sigma \mid \lambda \sim IW(\Psi, d), \tag{A.10}$$

where $IW(\Psi, d)$ denotes an inverse-Wishart distribution with scale matrix Ψ and degrees of freedom d. The prior mean B_0 is fixed. The random object is B. In particular, if c_0 denotes the intercept prior mean, the intercept implied by Equation (A.9) satisfies

$$c \mid \Sigma \sim N(c_0, V_c\Sigma). \tag{A.11}$$

Thus, V_c controls the prior variance of the intercept. A large value of V_c makes the intercept prior diffuse relative to the prior on lag coefficients.

The matrix V_λ is diagonal. Its first entry is the intercept variance, and the remaining entries are the prior variances for the lag coefficients:

$$V_\lambda = \operatorname{diag}\left(\omega_0, \omega_{1,1}, \ldots, \omega_{n,1}, \omega_{1,2}, \ldots, \omega_{n,2}, \ldots, \omega_{1,p}, \ldots, \omega_{n,p}\right). \tag{A.12}$$

The intercept variance is

$$\omega_0 = V_c. \tag{A.13}$$

For the coefficient on lag ℓ of variable j, the prior variance component is

$$\omega_{j,\ell} = (d - n - 1)\lambda^2 \frac{1}{\ell^\alpha} \frac{1}{\psi_j}, \qquad j = 1, \ldots, n, \quad \ell = 1, \ldots, p. \tag{A.14}$$

The scalar λ is the global tightness parameter. Smaller values of λ shrink the lag coefficients more strongly toward their prior means, while larger values imply weaker shrinkage. The parameter α controls how quickly the prior variance declines with the lag length. The term ψ_j scales the prior variance for coefficients on variable j and is constructed from a univariate

innovation variance for that variable. d represents the degrees of freedom. Since the size of z_t is not observed, we fix it to one by normalizing the innovation variance of the latent equation.

Conditional on λ and the current latent path, the posterior is conjugate. Define

$$\overline{V}_\lambda = \left(X'X + V_\lambda^{-1}\right)^{-1}, \tag{A.15}$$

$$\overline{B}_\lambda = \overline{V}_\lambda \left(X'Y + V_\lambda^{-1} B_0\right), \tag{A.16}$$

$$\overline{S}_\lambda = \Psi + \left(Y - X\overline{B}_\lambda\right)' \left(Y - X\overline{B}_\lambda\right) + \left(\overline{B}_\lambda - B_0\right)' V_\lambda^{-1} \left(\overline{B}_\lambda - B_0\right). \tag{A.17}$$

The parameter draw is

$$\Sigma \mid Y, z, \lambda \sim IW(\overline{S}_\lambda, T + d), \tag{A.18}$$

$$\mathrm{vec}(B) \mid \Sigma, Y, z, \lambda \sim N\left(\mathrm{vec}(\overline{B}_\lambda),\, \Sigma \otimes \overline{V}_\lambda\right). \tag{A.19}$$

Following the GLP implementation, the marginal data density and the prior for λ jointly determine the posterior distribution of λ. The sampler updates λ with a Metropolis step. Conditional on the accepted value of λ, it draws Σ and B from Equations (A.18)–(A.19). After each covariance draw, the last row and last column of Σ are rescaled so that the innovation variance of the latent recession equation equals one. This probit-scale normalization is required because the binary recession indicator identifies the sign of z_t, but not its units.

A.2.2 Second block: drawing the latent recession factor

The latent recession path is drawn conditional on the VAR parameters. Each draw comes from a truncated normal distribution whose support is determined by the observed NBER recession classification: below zero in recession months and above zero in expansion months. The conditional mean and variance of this truncated normal distribution depend on the position of the date in the sample, because the information available for drawing z_t differs at the beginning, middle, and end of the sample. Therefore, implementation separates the dates into three groups: the first p observations, the interior observations, and the last p observations.

This separation is needed because z_t enters the VAR in two ways. At date t, it is part of the dependent vector y_t. At later dates, it appears as a lagged regressor. For an interior date, the likelihood terms involving z_t are available on both sides of t. At the beginning of the sample, the required lag history is incomplete. At the end of the sample, the future equations that would use z_t as a lagged regressor are not all observed. The sampler uses the interior truncated-normal draw when the full local likelihood is available and boundary

adjustments otherwise.

For the interior observations ($t = p+1, \ldots, T-p$), collect all Gaussian likelihood terms that depend on y_t. These terms come from the equation dated t, where y_t is the dependent vector, and from the equations dated $t+1, \ldots, t+p$, where y_t enters as a lagged regressor. The resulting conditional Gaussian kernel is

$$p(y_t \mid Y_{-t}, B, \Sigma) \propto \exp\left[-\frac{1}{2}(y_t - m_t)' P (y_t - m_t)\right], \tag{A.20}$$

with precision matrix

$$P = \Sigma^{-1} + \sum_{\ell=1}^{p} B_\ell' \Sigma^{-1} B_\ell. \tag{A.21}$$

The vector m_t is the conditional mean of y_t implied by the current values of c, $B_1, \ldots, B_p$, Σ, and the observations that enter the VAR equations involving y_t. Partition $y_t = (x_t', z_t)'$, $m_t = (m_{x,t}', m_{z,t})'$, and

$$P = \begin{pmatrix} P_{xx} & P_{xz} \\ P_{zx} & P_{zz} \end{pmatrix}. \tag{A.22}$$

The untruncated conditional distribution of the latent component is

$$z_t \mid Y_{-t}, B, \Sigma \sim N\left(m_{z,t} - P_{zz}^{-1} P_{zx}(x_t - m_{x,t}),\ P_{zz}^{-1}\right). \tag{A.23}$$

Denote this conditional mean and variance by $\mu_{z,t}$ and $\sigma_{z,t}^2$. The sampled value is drawn from the corresponding truncated normal distribution:

$$z_t \sim \begin{cases} N(\mu_{z,t}, \sigma_{z,t}^2) \text{ truncated to } (-\infty, 0), & S_t = 1, \\ N(\mu_{z,t}, \sigma_{z,t}^2) \text{ truncated to } (0, \infty), & S_t = 0. \end{cases} \tag{A.24}$$

For the first p observations, the implementation uses a Metropolis-Hastings boundary step. The proposal is a truncated normal draw that respects the sign implied by the observed recession classification. Since the complete interior likelihood is not available at the beginning of the sample, the proposal is accepted or rejected using the local Gaussian VAR likelihood implied by Equation (A.4).

For the last p observations, the sampler computes the conditional mean of the latent recession factor from the VAR forecast implied by the current values of c, $B_1, \ldots, B_p$, and Σ, together with the available observations. The draw is then taken from the corresponding truncated normal distribution.

A.2.3 MCMC algorithm

The reduced-form sampler can be summarized as follows:

1. Initialize the latent path $z^{(0)}$ using the observed recession classification, so that $z_t^{(0)} < 0$ in recession months and $z_t^{(0)} > 0$ in expansion months. Initialize λ around its posterior-mode value from the marginal data density.

2. Conditional on $z^{(i-1)}$, update $\lambda^{(i)}$ with a random-walk Metropolis step. Given $\lambda^{(i)}$, draw $\Sigma^{(i)}$ and $B^{(i)}$ from the conjugate posterior in Equations (A.18)–(A.19).

3. Normalize the covariance matrix so that the innovation variance of the latent recession equation equals one.

4. Conditional on $(B^{(i)}, \Sigma^{(i)})$, draw $z^{(i)}$ from the truncated-normal latent-factor sampler. Use the Metropolis-Hastings boundary step for the first p observations, the interior truncated-normal draw for $t = p+1, \ldots, T-p$, and the conditional-forecast boundary step for the last p observations.

5. Rebuild the augmented data matrix using $z^{(i)}$ and repeat Steps 2–4 until burn-in is complete and the desired number of posterior draws has been saved.

After burn-in, the implementation saves posterior draws of B, Σ, the reduced-form residuals, and the latent recession path. These reduced-form draws are the inputs to the structural-identification and scenario-analysis steps.

For each posterior draw and forecast origin, the model implies a Gaussian forecast distribution for future values of the latent recession factor. Let $\widehat{z}_{t+h}^{u}$ and $(\widehat{\sigma}_{t+h}^{z,u})^2$ denote the mean and variance of the unconditional forecast of z_{t+h}. Given the sign convention in Equation (A.3), the implied recession probability is

$$p_{t+h}^{u} = \Pr_t(S_{t+h} = 1) = \Phi\left(-\frac{\widehat{z}_{t+h}^{u}}{\widehat{\sigma}_{t+h}^{z,u}}\right). \tag{A.25}$$

Under a policy scenario, the mean is replaced by the scenario mean $\widehat{z}_{j,t+h}^{c}$ while the standard deviation is held at $\widehat{\sigma}_{t+h}^{z,u}$, which gives Equation (11).

A.3 Structural Identification

Here we describe how sign restrictions, narrative restrictions, contribution restrictions, and exact zero restrictions are imposed in the structural impact matrix A_0.

For each reduced-form draw, the reduced-form residual satisfies

$$u_t \sim N(0, \Sigma). \tag{A.26}$$

The structural representation is

$$u_t = A_0 \varepsilon_t, \qquad \mathbb{E}(\varepsilon_t \varepsilon_t') = I, \qquad A_0 A_0' = \Sigma. \tag{A.27}$$

Let C be a square root of Σ, so that $CC' = \Sigma$. Any admissible structural impact matrix can then be written as

$$A_0 = CQ, \tag{A.28}$$

where Q is an orthonormal matrix. Structural identification is therefore a problem of finding rotations Q that satisfy the imposed restrictions. The reduced-form draw fixes Σ and the VAR dynamics; the rotation determines the structural interpretation of the innovations.

A.3.1 Zero restrictions

Zero restrictions are imposed directly on the construction of Q. They are therefore exact restrictions on A_0, not inequalities checked after drawing an unrestricted rotation.

Let q_j denote column j of Q. An impact zero restriction for variable i and shock j requires

$$e_i' A_0 e_j = 0. \tag{A.29}$$

Using $A_0 = CQ$, this restriction becomes

$$e_i' C q_j = 0. \tag{A.30}$$

Therefore, requiring the impact response of variable i to shock j to be zero is equivalent to requiring q_j to be orthogonal to that row of C.

If shock j has several impact-zero restrictions, let $\mathcal{Z}_j^0$ denote the set of variables whose impact responses to shock j must be zero. For each $i \in \mathcal{Z}_j^0$, Equation (A.30) gives one linear restriction on q_j. Stacking these row restrictions gives

$$H_j^0 = \begin{bmatrix} e_{i_1}' C \\ e_{i_2}' C \\ \vdots \\ e_{i_m}' C \end{bmatrix}, \qquad i_1, \ldots, i_m \in \mathcal{Z}_j^0. \tag{A.31}$$

The impact-zero restrictions for shock j can then be written as

$$H_j^0 q_j = 0. \tag{A.32}$$

Thus, an admissible column q_j must lie in the null space of H_j^0. This null-space representation is useful because any vector drawn from that null space automatically satisfies all impact-zero restrictions for shock j.

When more than one shock has zero restrictions, the zero-restricted columns of Q must also remain mutually orthogonal. This is handled sequentially. Suppose some columns have already been drawn and collected in

$$Q_F = \begin{bmatrix} q_1 & \cdots & q_{j-1} \end{bmatrix}. \tag{A.33}$$

The next zero-restricted column q_j must satisfy two requirements. First, it must satisfy the zero restrictions, $H_j^0 q_j = 0$. Second, it must be orthogonal to the previously fixed columns, $Q_F' q_j = 0$. These two sets of linear restrictions can be combined as

$$\widetilde{H}_j = \begin{bmatrix} H_j^0 \\ Q_F' \end{bmatrix}, \qquad \widetilde{H}_j q_j = 0. \tag{A.34}$$

The augmented matrix $\widetilde{H}_j$ therefore collects all linear restrictions that the new column must satisfy: the zero restrictions from the economic identification scheme and the orthogonality restrictions required for Q to remain a rotation matrix.

Let N_j be an orthonormal basis for the null space of $\widetilde{H}_j$. For any vector a, any vector of the form $N_j a$ satisfies $\widetilde{H}_j N_j a = 0$. We draw a random direction inside this admissible subspace by setting

$$q_j = \frac{N_j z}{\|N_j z\|}, \qquad z \sim N(0, I). \tag{A.35}$$

The normalization makes q_j a unit vector. After all zero-restricted columns are drawn, the remaining columns of Q are drawn from the orthogonal complement of the fixed columns.

A.3.2 Sign restrictions

The sign restrictions are imposed on impulse responses generated by A_0. Let $\mathcal{B}$ denote the companion matrix associated with the reduced-form VAR, and let J select the contemporaneous variables from the companion-form vector. The impulse response at horizon h is

$$\Theta_h = J \mathcal{B}^h J' A_0, \qquad h = 0, 1, \ldots, H. \tag{A.36}$$

A sign restriction on variable i, shock j, and horizon h can be written as

$$s_{i,j,h}\, e_i' \Theta_h e_j > 0, \tag{A.37}$$

where $s_{i,j,h} \in \{-1, 1\}$ gives the required sign.

These restrictions are set identifying. They do not choose a unique A_0. For each reduced-form draw, the algorithm draws candidate rotations and retains those whose impulse responses satisfy the sign restrictions.

A.3.3 Narrative sign and contribution restrictions

After a candidate rotation satisfies the zero and sign restrictions, the algorithm recovers the structural shocks from the reduced-form residuals:

$$\varepsilon_t = A_0^{-1} u_t = (CQ)^{-1} u_t. \tag{A.38}$$

Narrative sign restrictions are then imposed on the recovered shocks. For a narrative date τ and shock j, the restriction is

$$s_j^N\, \varepsilon_{\tau,j} > 0, \tag{A.39}$$

where s_j^N is the required narrative sign.

The contribution restriction uses the historical contribution of each structural shock to a selected variable over a specified window. Let the window be $\tau_0, \ldots, \tau_1$ and let $L_\tau = \tau_1 - \tau_0 + 1$. The contribution of shock j to variable i over that window is

$$\kappa_{i,j}(\tau_0, \tau_1; Q) = \sum_{h=0}^{L_\tau - 1} e_i' \Theta_h e_j\, \varepsilon_{\tau_1 - h, j}. \tag{A.40}$$

For a one-month window, this reduces to the impact contribution of shock j to the reduced-form innovation of variable i at the narrative date. The contribution restriction used in the implementation requires the target shock to be the largest absolute contributor:

$$\left|\kappa_{i,j^*}(\tau_0, \tau_1; Q)\right| \geq \max_{1 \leq j \leq n} \left|\kappa_{i,j}(\tau_0, \tau_1; Q)\right|. \tag{A.41}$$

This restriction follows the narrative-sign-restriction logic of Antolín-Díaz and Rubio-Ramírez (2018): the historical episode is used to discipline which structural innovation is allowed to explain the relevant reduced-form movement.

A.3.4 Structural-identification algorithm

The structural-identification step can be summarized as follows:

1. For each reduced-form posterior draw, compute a square root C of Σ and write candidate structural impact matrices as $A_0 = CQ$.

2. **Zero Restriction:** Construct the zero-restricted columns of Q. For each zero-restricted shock, stack the relevant rows of C and the orthogonality restrictions from already fixed columns, draw the new column from the null space of the resulting matrix, and normalize it to have unit length. Complete the remaining columns of Q from the orthogonal complement of the fixed columns.

3. **Sign Restriction:** Compute the impulse responses Θ_h using the companion matrix associated with $B_1, \ldots, B_p$. Apply the sign restrictions to the impulse responses.

4. **Narrative Restriction:** Recover the structural shocks $\varepsilon_t = (CQ)^{-1}u_t$ and evaluate the narrative sign and contribution restrictions.

5. Retain the draw only if sign and narrative restrictions hold until the desired number of accepted structural draws is reached.

These accepted structural draws are the inputs to the scenario analysis, which uses A_0 and the VAR dynamics to map future structural shocks into future macroeconomic paths.

A.4 Minimum-Effort Scenario Solution

A.4.1 Stacked structural representation

For a given posterior draw, let Θ_h denote the structural impulse responses in Equation (A.36). Consider an episode dated t, with an information set that includes data through $t-1$. Let $\widehat{y}^u_{t+h}$ denote the unconditional forecast at horizon h, obtained by setting future structural shocks to zero, and let $\widehat{y}^c_{t+h}$ denote the scenario forecast. Iterating the identified SVAR forward gives

$$\widehat{y}^c_{t+h} = \widehat{y}^u_{t+h} + \sum_{\ell=0}^{h} \Theta_{h-\ell}\widehat{\varepsilon}^c_{t+\ell}, \qquad h = 0, \ldots, H. \tag{A.42}$$

Stacking horizons $0, \ldots, H$ gives

$$\widehat{\mathbf{y}}^c_{t,H} = \widehat{\mathbf{y}}^u_{t,H} + M_H \widehat{\boldsymbol{\varepsilon}}^c_{t,H}, \tag{A.43}$$

where $\widehat{\mathbf{y}}^c_{t,H}$ and $\widehat{\boldsymbol{\varepsilon}}^c_{t,H}$ stack $\widehat{y}^c_{t+h}$ and $\widehat{\varepsilon}^c_{t+h}$ over $h = 0, \ldots, H$, and

$$M_H = \begin{bmatrix} \Theta_0 & 0 & \cdots & 0 \\ \Theta_1 & \Theta_0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \Theta_H & \Theta_{H-1} & \cdots & \Theta_0 \end{bmatrix}. \tag{A.44}$$

M_H maps a future structural-shock path into deviations of the scenario forecast from the unconditional forecast.

The full shock path has $n(H+1)$ entries. Under policy specification j, only the available policy shocks can be nonzero, and $M_{j,H}$ retains the columns of M_H that correspond to these shocks, which gives Equation (7). The number of free shock entries is $H+1$ when one policy shock can adjust and $2(H+1)$ when both can. In the benchmark, $n = 6$ and $H = 48$, so the full path has 294 entries and the free entries number 49 or 98.

A.4.2 Structural scenario analysis and restriction counting

Structural scenario analysis (Antolín-Díaz et al., 2021) defines a scenario through linear equality restrictions on future endogenous variables and on structural shocks. Let L select the variable–horizon pairs fixed by the scenario and let $\mathbf{f}_{t,H}$ collect their target values. Let Ξ select the shock–horizon entries fixed by the scenario and let $\mathbf{g}_{t,H}$ collect their imposed values. The restrictions are

$$L\widehat{\mathbf{y}}^c_{t,H} = \mathbf{f}_{t,H}, \qquad \Xi\widehat{\boldsymbol{\varepsilon}}^c_{t,H} = \mathbf{g}_{t,H}. \tag{A.45}$$

Restrictions are counted entry by entry. Each variable–horizon pair adds one row to L, so restricting inflation at 25 horizons contributes 25 restrictions. Each shock–horizon entry adds one row to Ξ, so setting one structural shock to zero over 49 months contributes 49 restrictions. Let k_y and k_ε denote the numbers of rows of L and Ξ.

Substituting Equation (A.43) expresses both sets of restrictions in shock space:

$$D_{t,H}\widehat{\boldsymbol{\varepsilon}}^c_{t,H} = \mathbf{b}_{t,H}, \qquad D_{t,H} = \begin{bmatrix} LM_H \\ \Xi \end{bmatrix}, \qquad \mathbf{b}_{t,H} = \begin{bmatrix} \mathbf{f}_{t,H} - L\widehat{\mathbf{y}}^u_{t,H} \\ \mathbf{g}_{t,H} \end{bmatrix}. \tag{A.46}$$

Each row of $D_{t,H}$ is one scalar condition imposed by the scenario, and each column corresponds to one structural-shock entry. Structural scenario analysis requires $D_{t,H}$ to have full

row rank, which in turn requires

$$k_y + k_\varepsilon \leq n(H+1). \tag{A.47}$$

When the restriction matrix has full row rank, which requires Condition (A.47) to hold, the scenario shock path is

$$\widehat{\boldsymbol{\varepsilon}}^c_{t,H} = D^\dagger_{t,H}\mathbf{b}_{t,H}, \qquad D^\dagger_{t,H} = D'_{t,H}(D_{t,H}D'_{t,H})^{-1}, \tag{A.48}$$

the path with the smallest sum of squared structural shocks among all paths that satisfy the restrictions exactly (Penrose, 1956). Because the structural shocks are independent standard normals, it is also the most likely shock path consistent with the scenario.[13] If $k_y + k_\varepsilon = n(H+1)$, the shock path is exactly determined by the restrictions. If $k_y + k_\varepsilon > n(H+1)$, the restrictions generically cannot all hold, and $D^\dagger_{t,H}\mathbf{b}_{t,H}$ returns a least-squares approximation that violates some of them.

A.4.3 Restriction counts with a recession-risk cap

Condition (A.47) compares the number of restrictions on endogenous variables, k_y, with the number of shock entries left free by the shock restrictions, $n(H+1) - k_\varepsilon$. The shock restrictions set the structural shocks not available to policy to zero at every horizon. Of the $n(H+1) = 294$ entries in the shock path, this leaves $H+1 = 49$ free entries in the single-policy specifications and $2(H+1) = 98$ in the dual-policy specification.

The restrictions on endogenous variables come from two sources. The inflation objective fixes annualized monthly inflation at Π^* for $h = T_\pi, \ldots, H$, which gives $H - T_\pi + 1 = 25$ equality restrictions in every specification. The recession-risk cap adds one inequality per horizon, and only the binding ones hold as equalities. Let $H^b \in \{0, \ldots, H+1\}$ denote the number of horizons at which the cap binds. Then

$$k_y = (H - T_\pi + 1) + H^b = 25 + H^b,$$

which ranges from 25 to 74 when binding recession-risk inequalities are treated as equalities. If the recession-risk cap were instead imposed as an equality at every horizon under structural scenario analysis, the number of equalities would be 74, corresponding to $H^b = H + 1 = 49$.

[13] Structural scenario analysis also delivers a conditional covariance for the shock path, $I - D^\dagger_{t,H}D_{t,H}$, which reduces forecast uncertainty under the scenario. We use the mean path only and hold the forecast standard deviation of the latent factor at its unconditional value, which keeps the recession-risk restriction linear in the shock path.

Table A.1 compares the number of equality restrictions with the free shock entries under the full-row-rank requirement of structural scenario analysis. The inflation objective alone satisfies the necessary restriction-counting condition in every specification. In the single-policy specifications, the 49 free shock entries accommodate at most 24 additional recession-risk equalities alongside the 25 inflation equalities. Imposing the cap as an equality at all 49 horizons would therefore violate the full-row-rank requirement. In the dual-policy specification, the 98 free entries are sufficient to accommodate all 74 variable equalities, provided the restrictions are linearly independent. Nevertheless, imposing equality at every horizon would require recession probability to equal the cap rather than remain at or below it. Importantly, these restriction counts characterize the applicability of the equality-based structural scenario formulation.

Table A.1: Restrictions and Free Shock Entries

	Free shock	Restrictions on endogenous variables, $k_y = 25 + H^b$		
Specification	entries $n(H+1) - k_\varepsilon$	Inflation only $H^b = 0$	Cap at all horizons $H^b = H + 1$	Maximum under row-rank count
MP-only or AMP-only	49	25	74	49 $(H^b = 24)$
Dual policy	98	25	74	74 $(H^b = 49)$

Notes: Condition (A.47) holds when k_y does not exceed the free shock entries. The VAR has $n = 6$ variables and the forecast horizon is $h = 0, \ldots, 48$, so the full shock path has $n(H+1) = 294$ entries. The shock restrictions set the shocks not available to policy to zero at every horizon, so $k_\varepsilon = 245$ in the single-policy specifications and $k_\varepsilon = 196$ in the dual-policy specification. The inflation objective contributes 25 restrictions, and H^b is the number of horizons at which the recession-risk bound in Equation (12) binds. "Cap at all horizons" imposes the bound as an equality at all 49 horizons, as structural scenario analysis would require. "Maximum under row-rank count" reports the maximum number of equalities consistent with the necessary restriction-counting condition, assuming that the relevant rows are linearly independent.

A.4.4 Minimum-effort problem

Equation (15) retains the shock restrictions, the inflation restrictions, and the minimum-norm criterion of structural scenario analysis and replaces the equality treatment of recession risk with an inequality. It imposes three sets of restrictions.

1. *Shock restrictions.* Only the policy shocks available under specification j enter $\boldsymbol{\varepsilon}_{j,t,H}$. This is the restriction $\Xi \boldsymbol{\varepsilon}_{t,H} = \mathbf{0}$ in Equation (A.45), implemented by retaining only the corresponding columns of M_H in $M_{j,H}$.

2. *Inflation restrictions.* The $H - T_\pi + 1$ equality restrictions in Equation (9).

3. *Recession-risk restrictions.* The $H + 1$ inequality restrictions in Equation (13). With $\bar{p} = 0.3$, $\Phi^{-1}(\bar{p}) \approx -0.524$, so the bound is $\widehat{z}^{\,c}_{j,t+h} \geq 0.524\, \widehat{\sigma}^{z,u}_{t+h}$.

The loss function $\frac{1}{2}\boldsymbol{\varepsilon}'_{j,t,H}\boldsymbol{\varepsilon}_{j,t,H}$ weights every shock entry equally, because the structural shocks are mutually independent with unit variance. Under normality of the structural shocks, the loss equals the negative log likelihood of the policy-shock path up to an additive constant. The minimizer is therefore the most likely policy-shock path that delivers both objectives. This is the same criterion that selects Equation (A.48) in structural scenario analysis.

The scenario parameters are $\Pi^* = 2$, $T_\pi = 24$, $H = 48$, and $\bar{p} = 0.3$. Section 3.2.4 and Appendix C.3 vary T_π and $\bar{p}$. For each posterior draw, the forecast standard deviation of the latent factor is computed from the unconditional forecast,

$$\widehat{\sigma}^{z,u}_{t+h} = \left(\sum_{\ell=0}^{h} e'_z \Theta_\ell \Theta'_\ell e_z \right)^{1/2}, \tag{A.49}$$

where e_z selects the latent factor, and is held fixed under the scenario.

Existence and uniqueness. The objective is strictly convex, and the constraints are linear in the shock path because $\widehat{\sigma}^{z,u}_{t+h}$ is held fixed. The set of admissible paths is therefore convex. Whenever the feasible set is nonempty, Equation (15) has a unique solution, and the Karush–Kuhn–Tucker conditions are necessary and sufficient. We solve the problem for each posterior draw with OSQP (Stellato et al., 2020), an operator-splitting solver for convex quadratic programs that also detects infeasible problems.

B Historical Starting Conditions

B.1 Historical Episode Comparison

Section 3.1 uses May 1981 and July 2022 as illustrations to describe the policy-design mechanism. For additional historical context, this appendix shows the same benchmark counterfactuals for six salient U.S. anti-inflation episodes. The dates are motivated by the narrative literature on major U.S. anti-inflation policy episodes (Romer and Romer, 1989, 1994, 2023). We also include February 1994, the beginning of the Federal Reserve's 1994–95 tightening cycle, which is widely regarded as a successful soft landing (Blinder, 2023).

Table B.1 shows that the patterns highlighted in May 1981 and July 2022 extend to the other historical salient episodes. October 1979, similar to May 1981, combines a large remaining disinflation requirement with an adverse recession outlook, while February 1994 represents a comparatively mild policy burden. Despite these differences, the minimum-effort inflation-only solutions have very similar compositions, with AMP accounting for about 30

Table B.1: Benchmark Counterfactuals across Historical Episodes

	1978:08	1979:10	1981:05	1988:12	1994:02	2022:07
A. Initial Policy Burden						
Initial 12-month CPI inflation (%)	7.4	11.2	9.7	4.2	2.4	8.6
Required policy disinflation at month 24 (pp)	3.1	4.3	4.8	2.6	1.0	2.2
Peak recession probability, unconditional forecast (%)	23.7	58.1	81.7	20.4	12.6	22.6
B. Inflation-Only Recession Risk						
Policy effort (RMS, s.d.)	0.59	0.80	0.90	0.50	0.22	0.40
AMP share of policy-induced disinflation (%)	29.8	30.6	30.0	29.1	27.2	31.6
Peak recession probability (%)	67.6	93.1	98.0	54.7	19.5	48.3
C. Risk-Constrained Solution						
Policy effort (RMS, s.d.)	0.87	1.66	2.09	0.71	0.22	0.52
AMP share of policy-induced disinflation (%)	58.8	76.6	75.3	57.0	29.9	46.6

Notes: The table reports benchmark counterfactuals for six historically salient U.S. anti-inflation episodes. Required policy disinflation at month 24 is the reduction in annualized monthly CPI inflation required at month 24, relative to the unconditional forecast, to reach the 2 percent target. Peak recession probability is the maximum over months 0–48. Panel B reports the minimum-effort combination of MP and AMP shocks that satisfies the inflation objective without imposing the recession-risk cap. Panel C adds the 30 percent cap. Policy effort is the root-mean-square magnitude of available policy shocks over 0–48 months. The AMP share is the percentage of policy-induced disinflation attributable to AMP shocks. Entries are posterior medians.

percent of policy-induced disinflation across all six episodes. Imposing the recession-risk cap raises both policy effort and the AMP share, with the larger changes occurring in episodes where the inflation-only path generates particularly high recession risk.

B.2 Selection of Representative Starting Conditions

This subsection describes how the representative starting dates in Section 3.2 are selected. We characterize each eligible starting month using only three pre-scenario variables. For each month t, define the initial-condition vector

$$s_t^0 = \left(\pi_{t-1}^{12m} - \Pi^*,\, z_{t-1},\, r_{t-1}\right)', \tag{B.1}$$

where $\pi_t^{12m} = 100 \log(\mathrm{CPI}_t/\mathrm{CPI}_{t-12})$ is 12-month CPI inflation, $\Pi^* = 2\%$ is the inflation target, z_t is the posterior median latent recession factor, and r_t is the federal funds rate. Since recession occurs when $z_t < 0$, lower values of z_t indicate greater recession pressure.

The candidate sample runs from 1977:06 to 2022:07. A month is eligible if it is not already in recession in the previous month and if inflation is meaningfully above target:

$$\mathcal{E} = \left\{t : S_{t-1} = 0,\, \pi_{t-1}^{12m} - \Pi^* > 0.5\right\}. \tag{B.2}$$

This rule yields 284 eligible months out of 542 months.

We divide each variable into low, middle, and high groups using its 33rd- and 66th-percentile cutoffs. In raw units, the 12-month inflation-gap groups are

$$\begin{aligned} \text{Low inflation gap}: &\quad \pi_{t-1}^{12m} - \Pi^* \leq 1.101, \\ \text{middle inflation gap}: &\quad 1.101 < \pi_{t-1}^{12m} - \Pi^* \leq 2.200, \\ \text{High inflation gap}: &\quad \pi_{t-1}^{12m} - \Pi^* > 2.200. \end{aligned} \tag{B.3}$$

The latent-factor groups are

$$\begin{aligned} \text{Low latent factor}: &\quad z_{t-1} \leq 3.255, \\ \text{middle latent factor}: &\quad 3.255 < z_{t-1} \leq 4.510, \\ \text{High latent factor}: &\quad z_{t-1} > 4.510. \end{aligned} \tag{B.4}$$

The federal-funds-rate groups are

$$\begin{aligned} \text{Low FFR}: &\quad r_{t-1} \leq 4.742, \\ \text{middle FFR}: &\quad 4.742 < r_{t-1} \leq 7.191, \\ \text{High FFR}: &\quad r_{t-1} > 7.191. \end{aligned} \tag{B.5}$$

Table B.2 reports the distribution of eligible months across the $3 \times 3 \times 3$ grid defined by inflation pressure, latent recession factor, and the federal funds rate.

Within each populated cell, we select the actual historical month closest to the vector of within-cell medians after standardizing three variables. Let $\mathcal{G}_m$ denote cell m, and let $\widetilde{X}_t$ and $\widetilde{X}_m^{med}$ denote the standardized initial-condition vector and the vector of within-cell medians, respectively. The selected month is

$$t_m^* = \arg\min_{t \in \mathcal{G}_m} \left\| \widetilde{X}_t - \widetilde{X}_m^{med} \right\|_2. \tag{B.6}$$

This rule selects a real historical month representative of its classification cell. The procedure selects 24 representative starting dates, one from each populated cell. Table B.3 reports the historical month selected from each cell. The three empty cells are low-inflation-gap, high-FFR cells with low or middle values of the latent recession factor and the high-inflation-gap, low-FFR cell with a high value of the latent recession factor.

Table B.2: Eligible Months by Classification Cell

		Latent recession factor		
FFR	Inflation-gap	Low	middle	High
Low	High	15	2	0
	middle	7	11	11
	Low	14	21	13
middle	High	4	7	7
	middle	8	11	12
	Low	5	15	24
High	High	24	16	22
	middle	17	10	6
	Low	0	0	2

Notes: The table reports the number of eligible months in each cell of the $3 \times 3 \times 3$ classification grid for the monthly VAR, six-lag specification. The classification variables are the 12-month CPI inflation gap, the posterior median latent recession factor, and the federal funds rate. Groups are defined by the 33rd- and 66th-percentile cutoffs within the eligible sample.

Table B.3: Representative Historical Months By Classification Cell

Inflation-gap	Latent-factor	FFR	Representative month	Cell count
Low	Low	Low	1992:09	14
		middle	1991:11	5
		High	Empty	0
	middle	Low	2018:07	21
		middle	1996:05	15
		High	Empty	0
	High	Low	2005:03	13
		middle	1995:04	24
		High	1983:10	2
middle	Low	Low	1992:11	7
		middle	1991:09	8
		High	1985:05	17
	middle	Low	2011:07	11
		middle	1987:06	11
		High	1988:09	10
	High	Low	2004:12	11
		middle	1988:03	12
		High	1988:11	6
High	Low	Low	2021:12	15
		middle	1991:07	4
		High	1989:08	24
	middle	Low	2021:06	2
		middle	1977:11	7
		High	1979:08	16
	High	Low	Empty	0
		middle	1977:09	7
		High	1978:10	22

Notes: Each representative month is the historical month closest to the vector of within-cell medians in standardized three-dimensional space. The three empty cells contain no eligible months.

C Robustness Checks

C.1 Pandemic-Time-Dummy Estimation

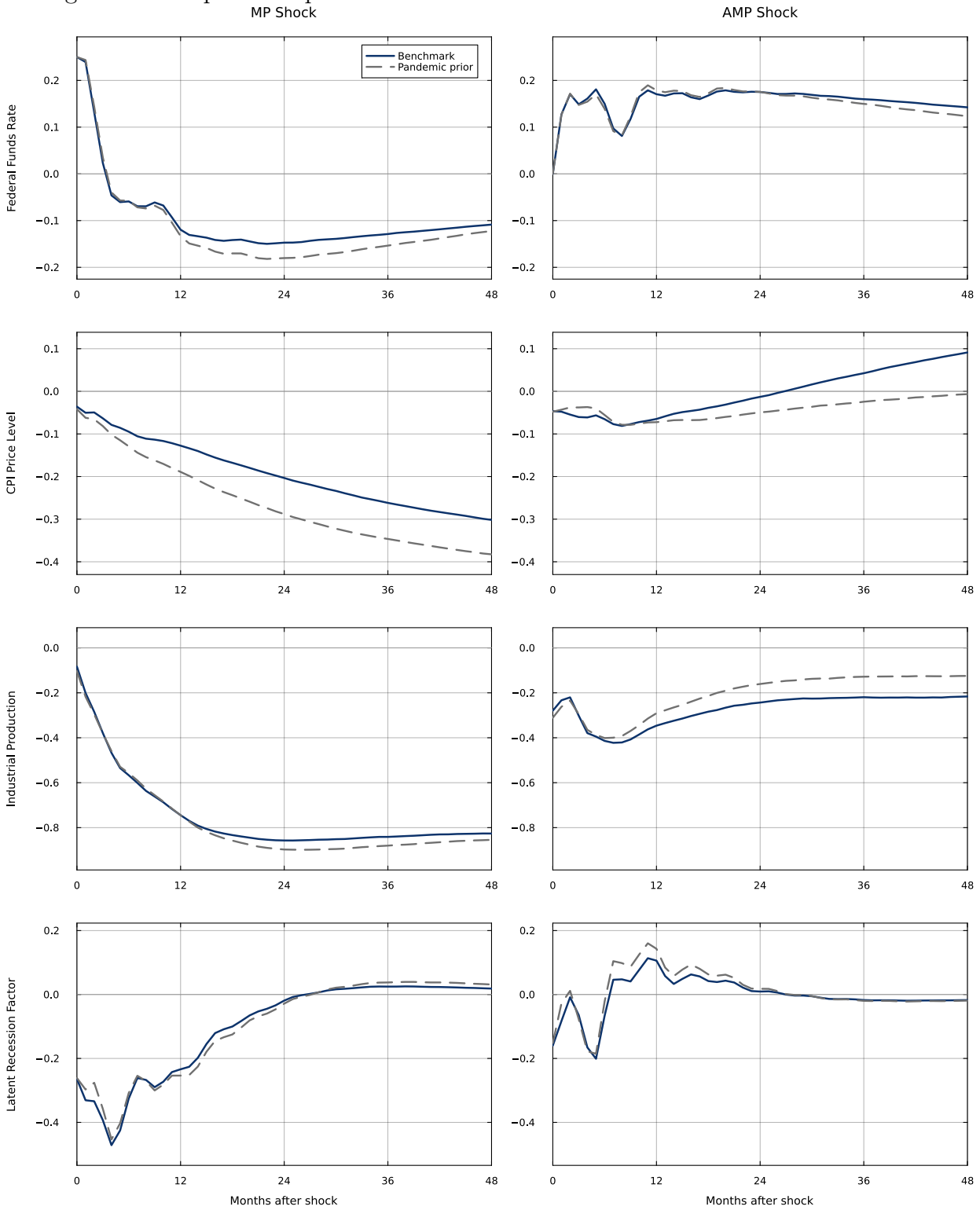
The benchmark Qual-VAR is estimated through December 2019. This raises a concern that the estimated reduced-form dynamics may not reflect relationships relevant to post-pandemic scenario origins.

To address this concern, we adapt the pandemic-time-dummy approach of Cascaldi-Garcia (2022) to the Qual-VAR. Specifically, we re-estimate the Qual-VAR through June 2022 and include six monthly dummies for March through August 2020 in every equation. Each dummy has an equation-specific coefficient, allowing pandemic-related shifts to differ across variables and months. All other aspects of the benchmark Qual-VAR are left unchanged.

Figure C.1 compares the estimated transmission of the two policy shocks. The distinction between the two policy shocks that is central to the scenario analysis is preserved. Under both specifications, the MP shock produces a substantially larger and more persistent contraction in industrial production and the latent recession factor than the AMP shock.

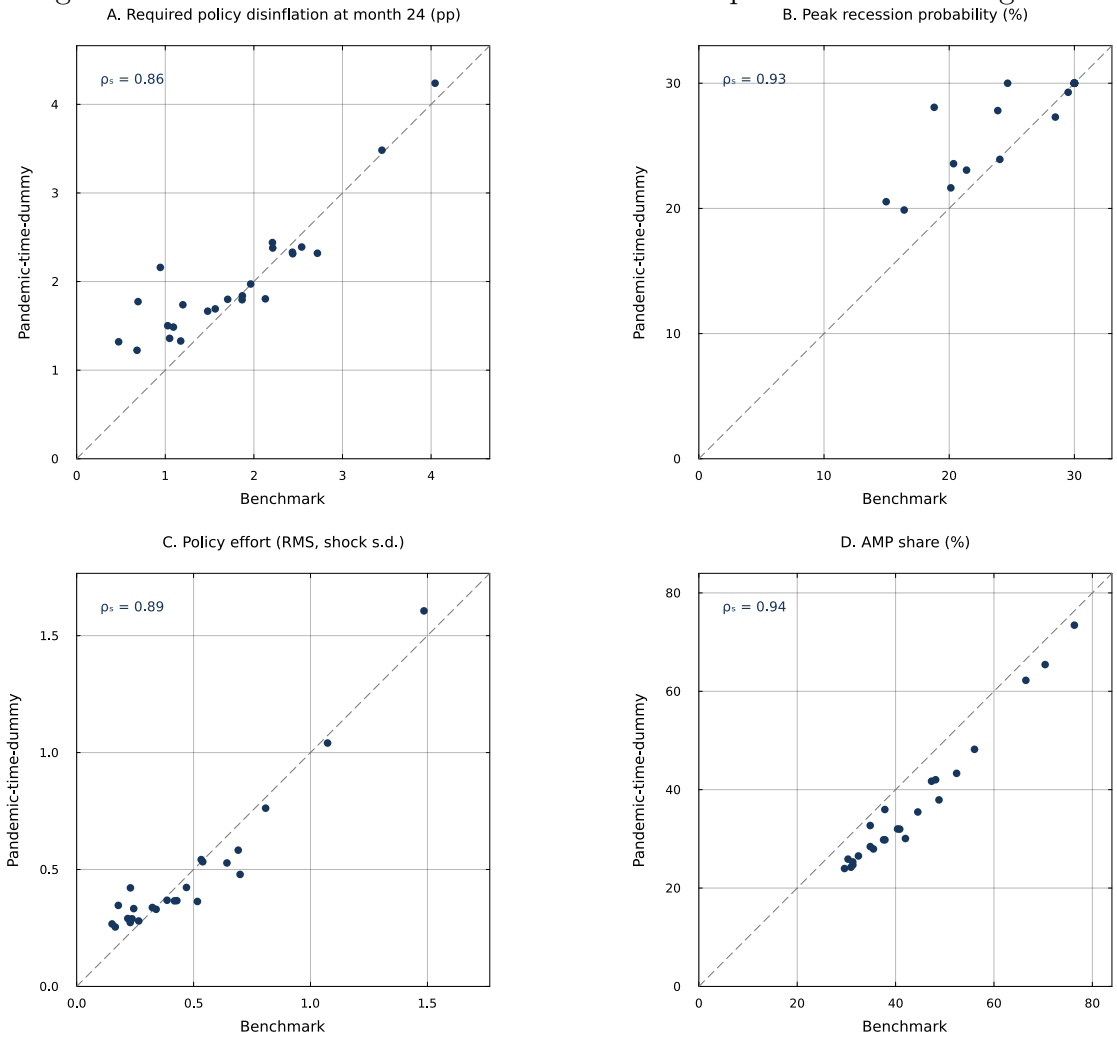
Figure C.2 compares the scenario outcomes across the same 24 representative starting dates. The rank ordering of the 24 starting dates is robust: the correlations range from 0.86 to 0.94 across the four outcomes.

Figure C.1: Impulse Responses under Benchmark and Pandemic-Period Estimation



Notes: Solid blue lines report posterior-median impulse responses under the benchmark Qual-VAR estimated through December 2019. Dashed gray lines report posterior medians under the Qual-VAR re-estimated through June 2022 using the pandemic-time-dummy specification described in the text. Columns distinguish MP and AMP shocks, while rows report the federal funds rate, the cumulative CPI price level, cumulative industrial production, and the latent recession factor.

Figure C.2: Pandemic-Period Robustness across Representative Starting Dates



Notes: Each point represents one of the 24 representative starting dates used in the benchmark analysis. The horizontal axis reports the outcome under the benchmark Qual-VAR estimated through December 2019, while the vertical axis reports the corresponding outcome under the Qual-VAR re-estimated through June 2022 using the pandemic-time-dummy specification. The dashed line is the 45-degree line. ρ_s denotes the Spearman rank correlation across the 24 starting dates.

C.2 Identification Robustness

This robustness check examines whether our results are sensitive to imposing a sharper distinction between the MP and AMP shocks. In particular, we consider a stronger identification in which the MP shock is restricted to have no contemporaneous effect on revisions to expected future policy rates. This restriction more sharply separates the MP shock, which moves the current federal funds rate on impact, from the AMP shock, which captures revisions to the anticipated future policy path.

To impose this restriction cleanly, we slightly modify the anticipated-policy indicator used in the baseline specification. The baseline indicator includes the revision to the current-quarter short-term interest rate together with revisions to expected future rates. Because the additional zero restriction is intended to exclude the MP shock specifically from revisions to future policy rates, we remove the current-quarter revision and define a future-only anticipated-policy indicator as

$$a_t^F = \frac{1}{3}\sum_{h=1}^{3} \Delta\widetilde{r}_{t|t+h}, \tag{C.1}$$

where $\Delta\widetilde{r}_{t|t+h}$ is the change between t and $t-1$ in the expected short-term interest rate at horizon h from the Blue Chip survey.

We then compare two identification specifications using this same future-only indicator. The first retains the baseline identifying restrictions and leaves the impact response of a_t^F to the MP shock unrestricted. The second additionally imposes that the MP shock has no impact effect on the future-only anticipated-policy indicator, a_0^F. Table C.1 summarizes the full set of restrictions under the two specifications.

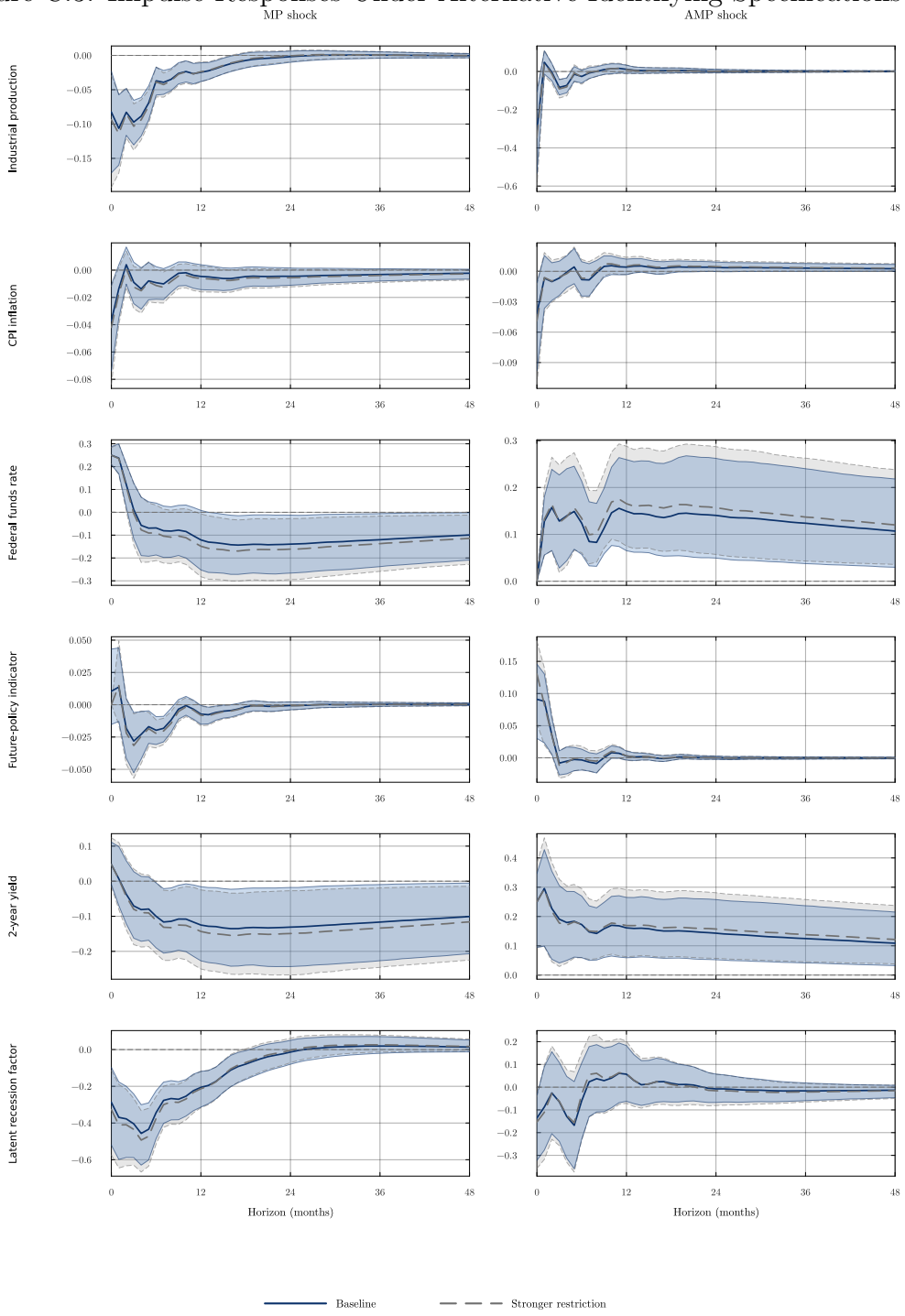
Figure C.3 compares the impulse responses under the two specifications. The solid blue lines report the posterior median impulse responses without the additional MP impact restriction, while the dotted gray lines report those under the stronger identification. The two specifications generate broadly similar response dynamics. In particular, the main distinction between the two shocks is preserved: the MP shock remains more contractionary for real activity and recession risk than the AMP shock.

Table C.1: Identification Restrictions in the Robustness Specifications

Shock	Δip_0	π_0	z_0	r_0	a_0^F	r_h	Narrative
Baseline							
MP	$-$	$-$	$-$	$+$			1979:10
AMP	$-$	$-$	$-$	0	$+$	$+$	
Stronger restriction							
MP	$-$	$-$	$-$	$+$	**0**		1979:10
AMP	$-$	$-$	$-$	0	$+$	$+$	

Notes: The table reports the identifying restrictions under the two robustness specifications. The variables are industrial production growth (Δip_t), CPI inflation (π_t), the latent recession factor (z_t), the federal funds rate (r_t), and the future-only anticipated-policy indicator (a_t^F). The entries $+$, $-$, and 0 denote positive, negative, and exact zero restrictions, respectively; an empty cell indicates that no restriction is imposed. The column r_h refers to the federal funds rate response at horizons $h = 6, \ldots, 18$ months. The narrative restriction applies to the MP shock and uses the October 1979 tightening episode.

Figure C.3: Impulse Responses Under Alternative Identifying Specifications

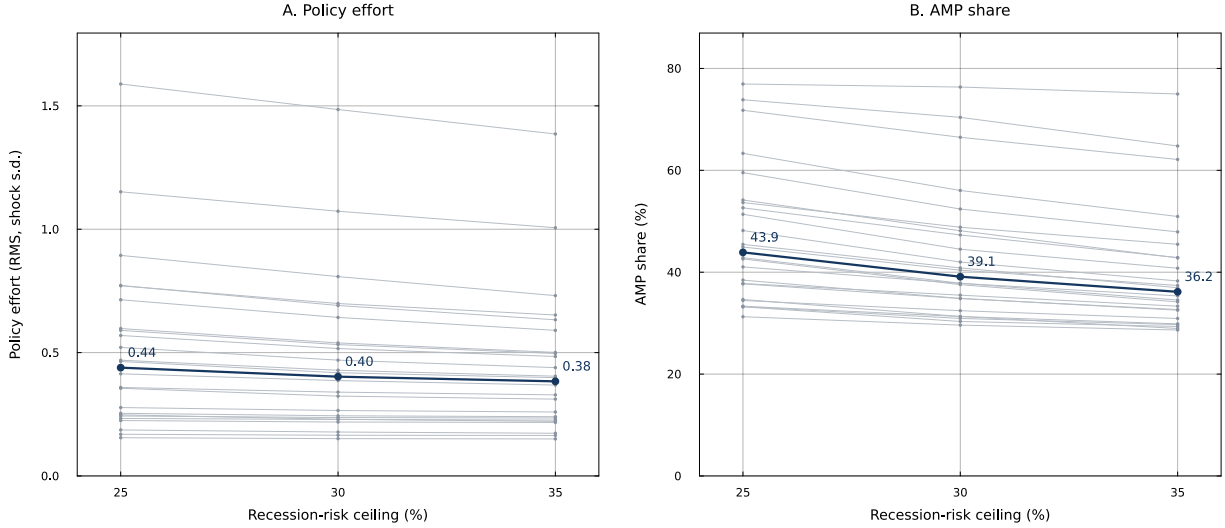


Notes: Solid blue lines and dotted gray lines report the posterior median impulse responses under the baseline and stronger-identification specifications, respectively. Shaded regions report the corresponding 68 percent credible sets. Columns distinguish MP and AMP shocks, while rows show variable responses. MP shock is normalized to raise the federal funds rate by 25 basis points on impact. AMP shock is normalized to raise the 2-year treasury yield by 25 basis points on impact.

C.3 Alternative Recession-Risk Cap

This appendix checks whether the results are robust to our 30 percent recession-risk cap. We compute the scenarios using a tighter ceiling of 25 percent and a looser ceiling of 35 percent.

Figure C.4: Robustness to Alternative Recession-Risk Caps



Notes: The figure shows policy effort and the AMP share under recession-risk caps of 25, 30, and 35 percent across the 24 representative starting dates. The blue line reports the median across starting dates.

Figure C.4 shows a clear and consistent pattern across the representative starting dates. Lowering the ceiling to 25 percent generally requires more policy effort and increases the AMP share, while raising it to 35 percent reduces both. These patterns hold broadly across starting dates. This preserves the main implication that tighter recession-risk constraints shift the policy mix toward AMP.